Hyperspectral Image Restoration via Global L_{1-2} Spatial–Spectral Total Variation Regularized Local Low-Rank Tensor Recovery

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***Abstract*— Hyperspectral images (HSIs) are usually corrupted by various noises, e.g., Gaussian noise, impulse noise, stripes, dead lines, and many others. In this article, motivated by the good performance of the L_{1-2} nonconvex metric in image sparse structure exploitation, we first develop a 3-D L_{1-2} spatial–spectral total variation (L_{1-2}SSTV) regularization to globally represent the sparse prior in the gradient domain of HSIs. Then, we divide HSIs into local overlapping 3-D patches, and low-rank tensor recovery (LTR) is locally used to effectively separate the low-rank clean HSI patches from complex noise. The patchwise LTR can not only adapt to the local low-rank property of HSIs well but also significantly reduce the information loss caused by the global LTR. Finally, integrating the advantages of both the global L_{1-2}SSTV regularization and local LTR model, we propose a L_{1-2}SSTV regularized local LTR model for hyperspectral restoration. In the framework of the alternating direction method of multipliers, the difference of convex algorithm, the split Bregman iteration method, and tensor singular value decomposition method are adopted to solve the proposed model efficiently. Simulated and real HSI experiments show that the proposed model can reduce the dependence on noise independent and identical distribution hypotheses, and simultaneously remove various types of noise, even structure-related noise.**

***Index Terms*— Hyperspectral images (HSIs), L_{1-2} spatial–spectral total variation (L_{1-2}SSTV), local low-rank tensor recovery (LTR), mixed noise, restoration.**

I. Introduction

HYPERSPECTRAL images (HSIs) are required by high spectral resolution sensors. They include hundreds of spectral bands ranging from ultraviolet to infrared wavelengths and can provide rich spatial–spectral information. Therefore, HSIs have drawn considerable attention from various application fields [1]. However, due to the intrinsic character and poor imaging system environment of the equipment, HSIs acquired by imaging sensors are inevitably corrupted by different noises, for example, Gaussian noise, impulse noise, deadlines, stripes, and their mixtures. The noise severely degrades the quality of the HSIs and further causes difficulties in the subsequent processing, including classification [2], mineral exploration [3], and target detection [4]. Therefore, as a preprocessing step to remove noise for various subsequent applications, HSI restoration is a valuable and active research topic.

To date, many restoration models have been proposed to reduce the noise in HSIs. A natural method is to utilize the traditional 1-D or 2-D denoising methods to restore HSIs pixel-by-pixel or band-by-band [5], [6], as each pixel of HSI can be regarded as a 1-D signal and each spectral band of HSI can also be regarded as a gray-level image. However, these denoising methods have not considered the correlations of all bands or spatial pixels in HSIs.

Consequently, to simultaneously use spatial and spectral information in HSIs and to improve the quality of restoration results, most researchers in recent literature have concentrated on extending 1-D or 2-D denoising models to 3-D models. Under the framework of Bayesian theory, Zhang *et al.* [7] averaged parts of spectral bands in an HSI to produce a latent multispectral image, and proposed a Bayesian restoration model based on the HSI and the latent multispectral image. Xu *et al.* [8] also proposed another Bayesian restoration model that used the spectral and spatial information in an HSI by the Monte Carlo sampling method. Based on the BM3D model [9], Maggioni *et al.* [10] first grouped the cubes of voxels in 3-D data and then applied collaborative filtering to denoise the different cube groups that were stacked into 4-D data. Moreover, Letexier and Bourennane [11] proposed a multidimensional Wiener filtering model by treating an HSI as a 3-D tensor cube, which jointly considered the spatial geometric structure and spectrum continuity. Because of combining the spatial and spectral prior information in HSIs, all of the above restoration models can achieve good performance in single-type noise or two-type noise removals such as, Gaussian noise, impulse noise, or hybrid Gaussian-impulse noise.

Actually, the observed HSIs are usually corrupted by complex types of noise, e.g., Gaussian noise, impulse noise, deadlines, stripes, and their mixtures [12]. The aforementioned denoising methods that only remove simple noise fail to depress complex mixed noise. Hence, to reduce the mixed

Manuscript received December 10, 2019; revised April 2, 2020 and May 23, 2020; accepted July 3, 2020. Date of publication July 20, 2020; date of current version March 25, 2021. This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant 2452019073, and in part by the National Natural Science Foundation of China under Grant 61876153. *(Corresponding author: Xiaozhen Xie.)*

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Digital Object Identifier 10.1109/TGRS.2020.3007945

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noise, two types of important prior information in HSIs, such as the low-rank property in the spectral domain and the sparse property in the spatial domain, are explored to refine the restoration methods. Based on the robust principal component analysis (RPCA) model [13], Zhang *et al.* [14] proposed an low-rank matrix recovery (LRMR) model. They lexicographically order 3-D HSIs into 2-D Casorati matrices whose columns comprise vectorized bands. Then, the low-rank prior is approximated by minimizing the nuclear norm of the Casorati matrix, and the sparse prior of noise in HSIs is represented by minimizing its L_1 norm. The final LRMR model can remove the sparse noise in HSIs. He *et al.* [15] combined an iterative regularization framework with the LRMR model and proposed a noise-adjusted iterative low-rank matrix approximation (NAILRMA) method. To preserve edges and suppress artifacts effectively, Xue *et al.* [16] introduced the nonlocal LRMR model and the hyper-Laplacian prior to encoding the spatial and spectral structured sparsity. To obtain a tighter approximation of the rank function, Xie *et al.* [17] replaced the nuclear norm in the LRMR model with a nonconvex weighted Schatten p-norm, and Chen *et al.* [18] presented a nonconvex γ-norm. Furthermore, to reduce computational cost and enhance flexibility, a low-rank matrix factorization scheme [19], as another way to represent the low-rank prior, was proposed to denoise the HSIs. Although the above models have considered the spectral and spatial corelationships, they need to convert high-dimensional HSIs into 2-D matrices by vectorizing each band. This strategy will lose useful multiorder structure information.

In recent years, the HSI has been treated as a third-order tensor, and many studies [20]–[22] have proven that restoration methods directly modeling tensors can achieve better results than those modeling the HSIs' matricization. Then, tensor recovery based on the tensor low-rank property, low-rank tensor recovery (LTR), attracts much attention. In [12], the intrinsic structures of the clean HSI are identified using Tucker decomposition of tensors. Xue *et al.* [23] proposed an HSI denoising model using nonlocal low-rank tensor decomposition, which uses the CANDECOMP/PARAFAC decomposition to implicitly describe the intrinsic prior. Xie *et al.* [24] propose a denoising model by applying the intrinsic tensor sparsity measure on tensors formed by nonlocal similar patches. Furthermore, Fan *et al.* [25] and Chen *et al.* [26] introduced the tensor singular value decomposition (t-SVD) method [27] into the RPCA model, and proposed the tensor RPCA (TRPCA) model to denoise images. In their model, the tensor tubal rank and tensor nuclear norm induced by t-SVD are used to represent the low-rank prior of images, obtaining high-quality restoration results. In addition, to implicitly describe the intrinsic prior, Xue *et al.* [28] proposed an enhanced sparsity prior model using both the local and the global sparsity information of low-rank tensors, and Xue *et al.* [29] presented a new model to approximate the tensor rank using the nonconvex logarithmic surrogate function with respect to the singular values.

TV regularization is another efficient tool in the image processing field. Due to its good performance in preserving the spatial piecewise smoothness and edge structures of images, spatial sparsity, it is widely used in the denoising task [30], magnetic resonance processing [31], image superresolution [32], and reconstruction [33]. First, TV regularization is introduced to denoise the gray-level image in the ROF model [34]. It can also be extended to process the color images, which is named color TV in [35]. Compared to the gray or color images, HSIs have hundreds of spectral bands. The distributions and variances of the noises in each band are different. Then, for HSI restoration problems, the LRTV model [36] uses TV regularization to preserve the spatial sparsity bands by bands. Chang *et al.* [37] extended the classic TV to the 3-D anisotropic spatial–spectral total variation (SSTV), which regularizes not only the spatial sparsity but also the spectral sparsity. Considering the difference in noise variances in different spectral bands and different spatial locations simultaneously, Yuan *et al.* [38] proposed spatial–spectral adaptive total variation (SSATV) regularization. Wang *et al.* [12] and Fan *et al.* [39] combined the 3-D total variation regularizer and tensor low-rank decomposition or approximation in a unified framework. Recently, Lou *et al.* [40] proposed the difference of anisotropic and isotropic total variation (L_{1-2}TV) regularization to more approximately represent the spatial sparsity in gray-level images. Xie *et al.* [41] substituted the classic TV for this new TV definition in the LRTV model and reach better restoration results.

Although the abovementioned HSI denoising methods show great success in dealing with various issues, there are still a few open questions that have yet to be addressed. First, although the classic TV of grayscale images has been extended to HSIs, its disadvantages have also been brought to HSIs, that is, the artifact phenomenon still exists. Second, HSI denoising methods based on Tucker, PARAFAC, and t-SVD decomposition are shown in [12], [25], [39], and [42] only employ the global correlation across spectrum prior information in their models and ignore the local low-rank prior, which is a very important prior in most HSI data sets.

To alleviate these problems, in this article, we propose a mixed noise removal model by combining local LRT and L_{1-2}SSTV regularization, named the TLR-L_{1-2}SSTV model. From one aspect, to explore the local low-rank prior, HSIs are divided into overlapping 3-D patches, each of which is supposed to lie in a low-dimensional subspace. The local LTR-based method can be used to efficiently separate the low-rank clean HSI patches from the sparse noise. However, this method does not make full use of the spatial sparse prior, so it cannot effectively remove heavy Gaussian noise. From another aspect, it is also assumed that the HSI is sparse in the gradient domain. To reduce artifacts, we propose a new 3-D L_{1-2}SSTV regularization to capture global gradient sparsity and remove Gaussian noise effectively. Inspired by the above facts, integrating local LTR with global L_{1-2}SSTV regularization, we propose the TLR-L_{1-2}SSTV model. The main contributions of this article are summarized as follows.

1) The 3-D L_{1-2}SSTV is proposed for HSI restoration. From the perspective of sparse representation, the L_{1-2} SSTV regularization can achieve a more approximate representation of the sparse prior in the gradient domain.

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Additionally, it can be used as a powerful regularization tool in other image processing fields.

2) A global L_{1-2}SSTV regularization is incorporated in the local LTR model. The tensor nuclear norm induced by t-SVD is used to approximate the rank function of local patches instead of the whole HSI. The proposed global L_{1-2}SSTV regularization is used to remove Gaussian noise and enhance the spatial information.
3) We design an effective algorithm to solve the model based on the difference of convex algorithm (DCA) method [43], the alternating direction method of multipliers (ADMM) [44], and the Bregman iteration method [45]. Furthermore, experimental results are reported to demonstrate the effectiveness of the proposed model and the efficiency of the numerical scheme.

The rest of this article is organized as follows. Section II introduces some notations and preliminaries of tensors. In Section III, the L_{1-2}SSTV regularized LTR model is proposed, and its optimization procedure is introduced. Both simulated and real HSI data experiments are carried out and analyzed in Section IV. Finally, we conclude this article in Section V.

II. Notation and Preliminaries

The order of a tensor is defined as the number of its dimensions. A real-valued tensor is denoted by capitalized calligraphic letters, for example, $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times \ldots \times I_n}$, and its $(i_1, i_2, \ldots, i_n)$-element is denoted by $a_{i_1,i_2,\ldots,i_n}$. In particular, an $n \times 1$ vector x is a first-order tensor, and an $n \times m$ matrix A can be regarded as a second-order tensor.

For a third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, let $A^{(i)} = \mathcal{A}(:, :, i)$ denote the ith frontal slice of $\mathcal{A}$. By utilizing the MATLAB command fft, we denote $\bar{\mathcal{A}}$ as the fast Fourier transform (FFT) along the third dimension, that is $\bar{\mathcal{A}} = \text{fft}(\mathcal{A}, [\,], 3)$. Similarly, we can obtain $\mathcal{A}$ by the inverse FFT, i.e., $\mathcal{A} = \text{ifft}(\bar{\mathcal{A}}, [\,], 3)$. $\bar{A}$ is a block diagonal matrix in which each block on diagonal is the frontal slice $\bar{A}^{(i)}$ of $\bar{\mathcal{A}}$, that is

$$\bar{A} = \text{bdiag}(\bar{\mathcal{A}}) = \begin{bmatrix} \bar{A}^{(1)} & & & \\ & \bar{A}^{(2)} & & \\ & & \ddots & \\ & & & \bar{A}^{(n_3)} \end{bmatrix}.$$

Additionally, the block circulant matrix $\text{bcirc}(\mathcal{A}) \in \mathbb{R}^{n_1 n_3 \times n_2 n_3}$ of $\mathcal{A}$ is defined as

$$\text{bcirc}(\mathcal{A}) = \begin{bmatrix} A^{(1)} & A^{(n_3)} & \cdots & A^{(2)} \\ A^{(2)} & A^{(1)} & \cdots & A^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ A^{(n_3)} & A^{(n_3-1)} & \cdots & A^{(1)} \end{bmatrix}.$$

In addition, as in [27], we define the following operator:

$$\text{unfold}(\mathcal{A}) = \begin{bmatrix} A^{(1)} \\ A^{(2)} \\ \vdots \\ A^{(n_3)} \end{bmatrix}, \quad \text{fold}(\text{unfold}\,\mathcal{A}) = \mathcal{A}.$$

Then, one can define the tensor–tensor product (t-product) [46] based on the block circulant matrix, the operators unfold and fold.

Definition 1 *(t-Product) [46]:* Let $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and $\mathcal{B} \in \mathbb{R}^{n_2 \times l \times n_3}$. Then, the t-product $\mathcal{A} * \mathcal{B}$ is defined to be a tensor of size $n_1 \times l \times n_3$

$$\mathcal{A} * \mathcal{B} = \text{fold}(\text{bcirc}(\mathcal{A}) \cdot \text{unfold}(\mathcal{B})).$$

Definition 2 *(Identity Tensor) [46]:* The identity tensor $\mathcal{I} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$, is defined as a tensor whose first frontal slice $I^{(1)}$ is the $n_1 \times n_1$ identity matrix, and all other frontal slices $I^{(i)}$, $i = 2, \ldots, n_3$ are zero.

Definition 3 *(Orthogonal Tensor) [46]:* A tensor $\mathcal{Q} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$, is orthogonal if it satisfies

$$\mathcal{Q}^\top * \mathcal{Q} = \mathcal{Q} * \mathcal{Q}^\top = \mathcal{I}.$$

Definition 4 *(f-Diagonal Tensor) [46]:* A tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is called the f-diagonal if each frontal slice $A^{(i)}$, $i = 1, 2, \ldots, n_3$ is a diagonal matrix.

Theorem 5 *(t-SVD) [46]:* Let $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ be a third-order tensor. Then, it can be factored as

$$\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$$

where $\mathcal{U} \in \mathbb{R}^{n_1 \times n_1 \times n_3}, \mathcal{V} \in \mathbb{R}^{n_2 \times n_2 \times n_3}$ are orthogonal, and $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f-diagonal tensor.

Then, one can compute the t-SVD based on the matrix SVD in the Fourier domain by performing the matrix SVD on the frontal slice of $\bar{\mathcal{A}}$, that is

$$\bar{A}^{(i)} = \bar{U}^{(i)} \bar{S}^{(i)} \bar{V}^{(i)*}$$

where $\bar{U}^{(i)}$, $\bar{S}^{(i)}$ and $\bar{V}^{(i)}$ are frontal slices of $\bar{\mathcal{U}}, \bar{\mathcal{S}}$, and $\bar{\mathcal{V}}$, respectively. Finally, by performing ifft along the third dimension, we can obtain the t-SVD $\mathcal{A} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$, where

$$\begin{aligned} \mathcal{U} &= \text{ifft}(\bar{\mathcal{U}}, [\,], 3) \\ \mathcal{S} &= \text{ifft}(\bar{\mathcal{S}}, [\,], 3) \\ \mathcal{V} &= \text{ifft}(\bar{\mathcal{V}}, [\,], 3). \end{aligned}$$

Definition 6 *(Tensor Nuclear Norm) [27]:* The tensor nuclear norm of $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, denoted as $\|\mathcal{A}\|_*$, is defined as the average of the nuclear norm of all the frontal slices of $\bar{\mathcal{A}}$, that is

$$\|\mathcal{A}\|_* := \frac{1}{n_3} \sum_{i=1}^{n_3} \left\| \bar{A}^{(i)} \right\|_*. \tag{1}$$

III. HSI Restoration via a Global L_{1-2}SSTV Regularized Local LTR

A. HSI Restoration Using LTR

The observed HSI is corrupted by mixed noise, which typically consists of Gaussian noise, stripes, impulse noise, deadlines, and so on [14], [15]. Let the third-order tensor $\mathcal{O} \in \mathbb{R}^{m \times n \times p}$ denote the observed HSI, where the spatial information lies in the first two dimensions and the spectral information lies in the third dimension. Then, the degradation model of the HSI can be formulated as

$$\mathcal{O} = \mathcal{L} + \mathcal{S} + \mathcal{N} \tag{2}$$

where $\mathcal{O}, \mathcal{L}, \mathcal{S}, \mathcal{N} \in \mathbb{R}^{m \times n \times p}$; $\mathcal{O}$ denotes the observed HSI; $\mathcal{L}$ represents the clean HSI data; $\mathcal{S}$ is the sparse noise, which

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consists of impulse noise, stripes, deadline, and so on; $\mathcal{N}$ is the Gaussian noise; $m \times n$ is the spatial size of the HSI, and p is the number of bands.

Under the framework of the degradation model (2), HSI restoration is a process of separating the mixed noise $\mathcal{S}, \mathcal{N}$ from the observed HSI $\mathcal{O}$, and restoring the clear HSI $\mathcal{L}$. In mathematical theory, this is a serious ill-posed problem. The regularization method is an effective and widely used method for solving such inverse problems. It establishes the following regularized restoration framework by adding prior information of the unknown clear HSI and mixed noise, that is:

$$\arg\min_{\mathcal{L}} \{\mathrm{J}(\mathcal{S}, \mathcal{N}) + \tau \mathrm{R}(\mathcal{L})\} \tag{3}$$

where $\mathrm{J}(\mathcal{S}, \mathcal{N})$ is a regular term to describe the distribution of different noises; $\mathrm{R}(\mathcal{L})$ is a regular term to represent the prior information of unknown clear HSI; τ is a nonnegative regularization parameter used to balance two regular terms. In the restoration framework (3), both prior information and the formulations of regular terms are important and determine the accuracy of the restoration results. Therefore, the research on HSI restoration mainly focuses on the exploration of prior information and the improvement of regularization formulations.

The TRPCA model [27], one famous method based on LTR, uses the minimization of the tensor nuclear norm to describe the low-rank property as follows:

$$\min_{\mathcal{L},S} \|\mathcal{L}\|_* + \lambda \|\mathcal{S}\|_1, \quad s.t. \quad \mathcal{O} = \mathcal{L} + \mathcal{S} \tag{4}$$

where $\|\cdot\|_*$ denotes the tensor nuclear norm defined in (1); $\|\cdot\|_1$ is the L_1 norm that detects the sparsity of sparse noise; The nonnegative parameter λ guarantees accurate recovery when $\mathcal{L}$ obeys the incoherence conditions, provided that the rank of $\mathcal{L}$ is not too large and S is reasonably sparse [47].

B. L_{1-2} SSTV Regularization

In recent years, TV regularization has been a promising technique for image processing. Because it can not only effectively protect the boundary information of the image, but also represent spatial smoothing information; it can guarantee the spatial sparsity in images [34]. TV has two mathematical definitions, namely, isotropic TV and anisotropic TV, as follows:

$$\begin{aligned} \|X\|_{\mathrm{TV}}^{\mathrm{iso}} &:= \sum_{i=1}^{mn} \sqrt{(D_x^i X)^2 + (D_y^i X)^2} \\ \|X\|_{\mathrm{TV}}^{\mathrm{ani}} &:= \sum_{i=1}^{mn} |D_x^i X| + |D_y^i X| \end{aligned} \tag{5}$$

where $X \in \mathbb{R}^{m \times n}$; D_x^i, D_y^i is the horizontal and vertical gradient operator at pixel i, respectively. Theoretical analysis [34], [48] has shown that these two definitions have different functions. Isotropic TV, the sum of squares of each gradient, which is suitable for describing Gaussian distribution can effectively detect and eliminate Gaussian random noise; While anisotropic TV, the sum of absolute value of each gradient, which is suitable for describing Laplace distribution can be used to eliminate pepper and impulse noise.

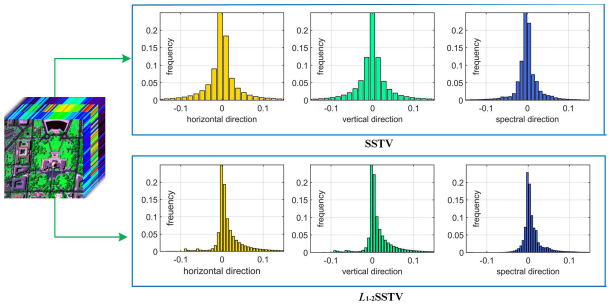

Fig. 1. Property of total variation of HSI in both spatial and spectral modes.

Compared to gray or color images, HSI data have hundreds of spectral bands. To make TV adapt to 3-D HSIs, subsequent researchers have proposed a large number of improved TV forms [49]–[51] which have been successfully applied to HSI restoration. Specifically, He *et al.* [52] extended 2-D TV and introduce 3-D SSTV of tensors as follows:

$$\|\mathcal{L}\|_{\mathrm{SSTV}}^{\mathrm{ani}} = \tau_x \|D_x \mathcal{L}\|_1 + \tau_y \|D_y \mathcal{L}\|_1 + \tau_z \|D_z \mathcal{L}\|_1 \tag{6}$$

where D_z denotes the gradient operator along the spectral dimension and τ_x, τ_y, τ_z are nonnegative regularization parameters. One can find an illustrative example for demonstrating the sparsity of the real HSI in Fig. 1 by different TV definitions. The left part of Fig. 1 lists real HSI data. In the right part of Fig. 1, the first row shows three frequency distributions, which are the difference values of gradient operators along the horizontal, vertical, and spectral dimensions. Not only in the spatial domain but also in the spectral dimension, it is obvious that most of the difference values are nearly equal to 0. Therefore, from the perspective of tensors, the SSTV regularization can fully consider the sparse prior in both spatial and spectral domains.

Actually, SSTV regularization, such as 3-D anisotropic TV regularization, uses the L_1 norm to measure the sparsity in the gradient domain. The L_1 norm is the optimal convex approximation of the L_0 norm, but it also has specific disadvantages. For example, the L_1 norm on the gradient, the anisotropic TV, produces stair-casing. The reason is that it prefers a piecewise constant image where the gradient at every pixel is 1-sparse. However, the images whose gradient vectors at the edges are more important may not be 1-sparse [40]. To promote the approximation of sparsity, several nonconvex forms, such as the $L_p, p \in (0, 1)$, $L_{1/2}$, and L_{1-2} norms [53]–[55] have been proposed to replace the L_1 norm. In particular, it has been proven that under certain conditions, the L_{1-2} norm is the best replacement among existing methods for recovering sparse solutions [53], [54], [56], it is Lipschitz continuous and can be minimized by the DCA method [43] where linearization convexifies the objective without additional smoothing or regularization. Briefly, the L_{1-2} norm can be written as the difference of the L_1 norm and the L_2 norm, that is

$$\|X\|_{L_{1-2}} := L_1 - \alpha L_2 = \|X\|_1 - \alpha \|X\|_2 \tag{7}$$

where $\alpha \in [0, 1]$ is a regularization parameter. When $\alpha = 0$, the norm is reduced to the L_1 norm. To better understand

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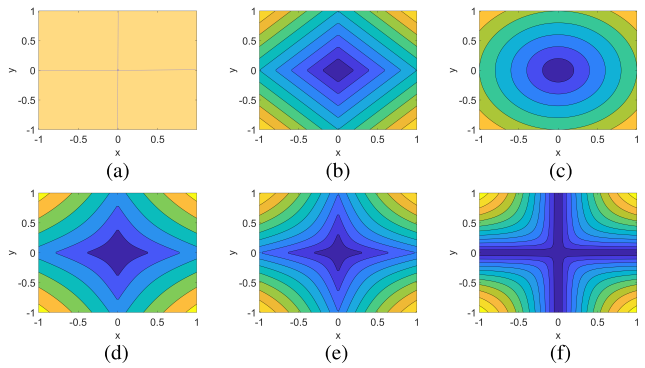

Fig. 2. Level curves of different sparse metrics. (a) L_0. (b) L_1. (c) L_2. (d) $L_1 - 0.5L_2$. (e) $L_1 - 0.7L_2$. (f) $L_1 - L_2$.

the character of the $L_1 - \alpha L_2$ metric, Fig. 2 lists the level curves of the L_0, L_1, L_2 norm, and L_{1-2} norms under the conditions of $\alpha = 0.5, 0.7, 1$. In Fig. 2(a), the value of the level curves of the L_0 norm at the origin is 0, at the axes is 1, and elsewhere is 2. Among the other level lines in Fig. 2(b)–(f), the level lines of the L_{1-2} norm when $\alpha \in (0, 1)$ are closest to those of the L_0 norm in the same condition. This means that L_{1-2} norm minimization has a more powerful ability for sparse representation.

Motivated by the above advantages of the SSTV and L_{1-2} norm, we substitute the L_1 norm in the SSTV for the L_{1-2} norm to obtain a more approximate representation of the sparse prior in the spatial and spectral domains simultaneously. Our proposed TV is named L_{1-2}SSTV and defined as follows:

$$\|\mathcal{L}\|_{L_{1-2}\text{SSTV}} = \|\mathcal{L}\|_{\text{SSTV}}^{\text{ani}} - \alpha\|\mathcal{L}\|_{\text{SSTV}}^{\text{iso}} \tag{8}$$

where $\alpha \in [0, 1]$ is a regularization parameter

$$\|\mathcal{L}\|_{\text{SSTV}}^{\text{iso}} = \sqrt{\|\tau_x D_x \mathcal{L}\|_1^2 + \|\tau_y D_y \mathcal{L}\|_1^2 + \|\tau_z D_z \mathcal{L}\|_1^2}.$$

In Fig. 1, the second row of the right part lists the three frequency distributions of the difference values of gradient operators along the horizontal, vertical, and spectral dimensions. Compared with the frequency distributions of SSTV in the three dimensions, the difference values of L_{1-2} SSTV are closer to 0. By combining isotropic TV and anisotropic TV, the proposed TV regularization can simultaneously provide a more approximate representation of sparse priors in the spatial and spectral domains, and thus is expected to be able to suppress mixed noise in an image more effectively.

The parameter α is an important parameter in definition (8). When $\alpha = 0$, L_{1-2}SSTV is reduced to SSTV. When $\tau_z = 0$, L_{1-2}SSTV is reduced to the 2-D L_{1-2} norm on the gradient. When α and τ_z are both equal to 0, L_{1-2}SSTV is reduced to 2-D anisotropic TV. Moreover, α, which is carefully estimated, can adjust the occurrence of nonsparse gradient vectors. Krishnan and Fergus [57] derive the value of α based on the gradient distribution. Lou *et al.* [40] provided a good choice $\alpha = 1/2$ based on experiments on the image gradient data.

C. Proposed HSI Restoration Model

In this section, we first define an operator $\mathcal{P}_{i,j} : \mathcal{L} \to \mathcal{L}_{i,j}$. This binary operator $\mathcal{P}_{i,j}$ is used to extract a 3-D patch $\mathcal{L}_{i,j} \in \mathbb{R}^{m_1 \times n_1 \times p}$ from the HSI data $\mathcal{L} \in \mathbb{R}^{m \times n \times p}$, where the spatial size of $m_1 \times n_1$ is centralized at pixel (i, j) of the HSI data. $\mathcal{P}_{i,j}^T$ is the inverse of $\mathcal{P}_{i,j}$. By the above definition, we apply the operation $\mathcal{P}_{i,j}$ on the tensors $\mathcal{O}$, $\mathcal{L}$, $\mathcal{S}$, $\mathcal{N}$, and divide them into overlapping patches as follows:

$$\mathcal{O}_{i,j} = \mathcal{P}_{i,j}(\mathcal{O}), \quad \mathcal{L}_{i,j} = \mathcal{P}_{i,j}(\mathcal{L})\mathcal{S}_{i,j} = \mathcal{P}_{i,j}(\mathcal{S})$$
$$\mathcal{N}_{i,j} = \mathcal{P}_{i,j}(\mathcal{N}), \quad (i, j) \in [1, m - m_1 + 1] \times [1, n - n_1 + 1].$$

Then, based on the TRPCA method [27] and above 3-D patches, the local LTR model can be described as

$$\begin{aligned} &\arg\min_{\mathcal{L},\mathcal{S}} \sum_{i,j} \|\mathcal{L}_{i,j}\|_* + \lambda\|\mathcal{S}_{i,j}\|_1 \\ &\text{s.t. } \|\mathcal{O}_{i,j} - \mathcal{L}_{i,j} - \mathcal{S}_{i,j}\|_F^2 \le \varepsilon. \end{aligned} \tag{9}$$

The local LTR model (9) can efficiently remove the sparse noise in the HSIs. This ability is directly inherited from the TRPCA model (4). Furthermore, its investigation of spectral similarity in a patchwise manner also has other advantages as follows. First, as mentioned in [52], pixels from the same local area in HSIs are more likely to be the same material, and the spectral signatures of the same material are more likely to be the same. Hence, it is more appropriate to explore the low-rank property in the local area rather than in the global area. Second, this patchwise approach can reduce the dependence on the hypothesis that noise in HSIs is independent and identically distributed (i.i.d.). Therefore, the local LTR model (9) has the ability to partially suppress the non-i.i.d. noises. Additionally, from the perspective of efficient algorithm implementation, all local patches can be treated globally and updated simultaneously by parallel computing technology.

Although the local LTR model (9) has great advantages in handling the more complex sparse noise, when the noise is structured or heavy, effective denoising still requires a full exploration of spatial information. To further explore the piecewise smoothness in both spatial and spectral domains, we incorporate L_{1-2}SSTV (8) into the local LTR model (9), and finally propose an L_{1-2}SSTV regularized local LTR (TLR-L_{1-2}SSTV) model for HSI restoration as follows:

$$\begin{aligned} &\arg\min_{\mathcal{L},\mathcal{S}} \sum_{i,j} (\|\mathcal{L}_{i,j}\|_* + \lambda\|\mathcal{S}_{i,j}\|_1) + \tau\|\mathcal{L}\|_{L_{1-2}\text{SSTV}} \\ &\text{s.t. } \|\mathcal{O}_{i,j} - \mathcal{L}_{i,j} - \mathcal{S}_{i,j}\|_F^2 \le \varepsilon \end{aligned} \tag{10}$$

where λ and τ are regularization parameters. It should be noted that in the proposed TLR-L_{1-2}SSTV model (10), the local LTR model is implemented in the local patches simultaneously to eliminate the sparse noises, even to partially suppress the non-i.i.d. noises. Additionally, the proposed L_{1-2}SSTV is globally used to reconstruct all the low-rank patches and efficiently preserve both the spatial and spectral sparsity of the whole HSI. With this strategy, the local and global regular terms are alternately implemented to remove complex mixed noises and restore clean HSIs.

D. Optimization Procedure

In this section, we use the ADMM to solve the TLR-L_{1-2}SSTV model (10). Auxiliary variables $\mathcal{J}, \mathcal{X} \in \mathbb{R}^{m \times n \times p}$

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are first introduced, and the TLR-L_{1-2}SSTV model (10) can be rewritten as follows:

$$\begin{aligned}\arg\min_{\mathcal{L},\mathcal{S},\mathcal{J},\mathcal{X}}\ &\sum_{i,j}(\|\mathcal{L}_{i,j}\|_*+\lambda\|\mathcal{S}_{i,j}\|_1)+\tau\|\mathcal{X}\|_{L_{1-2}\text{SSTV}}\\ &\text{s.t. } \mathcal{J}=\mathcal{X},\quad \mathcal{L}_{i,j}=\mathcal{J}_{i,j}\\ &\quad\|\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\|_F^2\le\varepsilon.\end{aligned}$$

By the ALM method, the above optimization model can be solved by minimizing the following augmented Lagrangian function:

$$\begin{aligned}&\arg\min_{\mathcal{L},\mathcal{S},\mathcal{J},\mathcal{X}}\ell(\mathcal{L},\mathcal{S},\mathcal{J},\mathcal{X})\\ &=\arg\min_{\mathcal{L},\mathcal{S},\mathcal{J},\mathcal{X}}\sum_{i,j}\\ &\quad\times\Big\{\big\|\mathcal{L}_{i,j}\big\|_*+\lambda\big\|\mathcal{S}_{i,j}\big\|_1+\langle\Lambda_{i,j}^{\mathcal{O}},\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\rangle\\ &\qquad+\frac{\mu}{2}\|\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\|_F^2\\ &\qquad+\langle\Lambda_{i,j}^{\mathcal{L}},\mathcal{L}_{i,j}-\mathcal{J}_{i,j}\rangle+\frac{\mu}{2}\big\|\mathcal{L}_{i,j}-\mathcal{J}_{i,j}\big\|_F^2\Big\}\\ &\quad+\langle\Lambda_{\mathcal{X}}\mathcal{J}-\mathcal{X}\rangle+\frac{\mu}{2}\|\mathcal{J}-\mathcal{X}\|_F^2+\tau\|\mathcal{X}\|_{L_{1-2}\text{SSTV}}\end{aligned}\tag{11}$$

where μ is the penalty parameter; $\Lambda_{i,j}^{\mathcal{O}}$, $\Lambda_{i,j}^{\mathcal{L}}$ and $\Lambda_{\mathcal{X}}$ are the Lagrangian multipliers; $\|\cdot\|_F^2$ represents the sum of squares of all elements in the tensor. The above minimization of the augmented Lagrangian function (11) can be solved by the ADMM method. In the kth iteration, its solution in the next iteration can be transformed into the following two subproblems:

$$(\mathcal{L}^{k+1},\mathcal{S}^{k+1})=\arg\min_{\mathcal{S},\mathcal{L}}\ell(\mathcal{L},\mathcal{S},\mathcal{J}^k)\tag{12}$$

$$(\mathcal{J}^{k+1},\mathcal{X}^{k+1})=\arg\min_{\mathcal{J},\mathcal{X}}\ell(\mathcal{L}^{k+1},\mathcal{J},\mathcal{X}).\tag{13}$$

Subproblem (12) can be regarded as the patchwise local LTR problem. Additionally, subproblem (13) is the globally L_{1-2}SSTV regularized reconstruction problem from the denoised patches in subproblem (12).

1) Local LTR Optimization for $(\mathcal{L},\mathcal{S})$: With the other parameters fixed, subproblem (12) for $(\mathcal{L},\mathcal{S})$ can be reformulated as

$$\begin{aligned}&\arg\min_{\mathcal{S},\mathcal{L}}\ell(\mathcal{L},\mathcal{S},\mathcal{J}^k)\\ &=\arg\min_{\mathcal{L}_{i,j},\mathcal{S}_{i,j}}\sum_{i,j}\big\|\mathcal{L}_{i,j}\big\|_*+\lambda\big\|\mathcal{S}_{i,j}\big\|_1\\ &\quad+\langle\Lambda_{i,j}^{\mathcal{O}},\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\rangle+\frac{\mu}{2}\|\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\|_F^2\\ &\quad+\langle\Lambda_{i,j}^{\mathcal{L}},\mathcal{L}_{i,j}-\mathcal{J}_{i,j}\rangle+\frac{\mu}{2}\big\|\mathcal{L}_{i,j}-\mathcal{J}_{i,j}\big\|_F^2.\end{aligned}$$

For the above optimization problem, we solve each patch separately and accumulate a weighted sum of $(\mathcal{L}_{i,j},\mathcal{S}_{i,j})$ to reconstruct $(\mathcal{L},\mathcal{S})$. With respect to each $(\mathcal{L}_{i,j},\mathcal{S}_{i,j})$, we have the following patchwise optimization problem:

$$\begin{aligned}&\arg\min_{\mathcal{L}_{i,j},\mathcal{S}_{i,j}}\|\mathcal{L}_{i,j}\|_*+\lambda\|\mathcal{S}_{i,j}\|_1\\ &+\langle\Lambda_{i,j}^{\mathcal{O}},\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\rangle+\frac{\mu}{2}\|\mathcal{O}_{i,j}-\mathcal{L}_{i,j}-\mathcal{S}_{i,j}\|_F^2\\ &+\langle\Lambda_{i,j}^{\mathcal{L}},\mathcal{L}_{i,j}-\mathcal{J}_{i,j}\rangle+\frac{\mu}{2}\|\mathcal{L}_{i,j}-\mathcal{J}_{i,j}\|_F^2.\end{aligned}\tag{14}$$

We alternately update the two variables $\mathcal{L}_{i,j}$ and $\mathcal{S}_{i,j}$. Then, the optimization problem (14) can be separated into two simpler minimization subproblems.

1) With $\mathcal{S}_{i,j}$ fixed, the minimization subproblem for $\mathcal{L}_{i,j}$ can be deduced from (14) as follows:

$$\begin{aligned}&\mathcal{L}_{i,j}^{k+1}\\ &=\arg\min_{\mathcal{L}_{i,j}}\|\mathcal{L}_{i,j}\|_*+2\\ &\quad\times\frac{\mu}{2}\left\|\mathcal{L}_{i,j}-\frac{1}{2}\big(\mathcal{O}_{i,j}+\mathcal{J}_{i,j}-\mathcal{S}_{i,j}+\big(\Lambda_{i,j}^{\mathcal{O}}-\Lambda_{i,j}^{\mathcal{L}}\big)/\mu\big)\right\|_F^2.\end{aligned}\tag{15}$$

The updating of $\mathcal{L}_{i,j}$ in (15) has a closed-form solution [25]. For the sake of simplicity, we denote the iteration of $\mathcal{L}_{i,j}$ in (15) as

$$\mathcal{L}_{i,j}^{k+1}=\arg\min_{\mathcal{L}_{i,j}}\|\mathcal{L}_{i,j}\|_*+2\times\frac{\mu}{2}\|\mathcal{L}_{i,j}-\mathcal{U}_{i,j}\|_F^2\tag{16}$$

where $\mathcal{U}_{i,j}=(1/2)(\mathcal{O}_{i,j}+\mathcal{J}_{i,j}-\mathcal{S}_{i,j}+(\Lambda_{i,j}^{\mathcal{O}}+\Lambda_{i,j}^{\mathcal{L}})/\mu)$. Solving the optimization subproblem (16) is equivalent to solving the following tensor recovery problem in the frequency domain:

$$\bar{\mathcal{L}}_{i,j}^{k+1}=\arg\min_{\bar{\mathcal{L}}_{i,j}}\frac{1}{p}\sum_{q=1}^{p}\big\|\bar{L}_{i,j}^{(q)}\big\|_*+2\times\frac{\mu}{2}\|\bar{\mathcal{L}}_{i,j}-\bar{\mathcal{U}}_{i,j}\|_F^2\tag{17}$$

where $\bar{\mathcal{L}}_{i,j}=\text{fft}(\mathcal{L}_{i,j},[\ ],3)$, $\bar{\mathcal{U}}_{i,j}=\text{fft}(\mathcal{U}_{i,j},[\ ],3)$, and $\bar{L}_{i,j}^{(q)}$ denotes the qth frontal slice of $\bar{\mathcal{L}}_{i,j}$. For the optimization subproblem in the frequency domain defined in (17), we can divide it into p independent minimization subproblems

$$\bar{L}_{i,j}^{k+1,(q)}=\arg\min_{\bar{L}_{i,j}^{(q)}}\big\|\bar{L}_{i,j}^{(q)}\big\|_*+2\times\frac{\mu}{2}\big\|\bar{L}_{i,j}^{(q)}-\bar{U}_{i,j}^{(q)}\big\|_F^2.\tag{18}$$

Lemma 1 *[58]:* Assume W is a matrix of size $m_1n_1\times p$, and the singular value decomposition of matrix W of rank r is

$$W=QE_rV^*,\ E_r=\text{diag}(\{\sigma_l\}_{1\le l\le r}).$$

The singular value shrinkage operator then obeys

$$\text{D}_\delta(W)=\arg\min_{\text{rank}(L)\le r}\delta\|L\|_*+\frac{1}{2}\|L-W\|_F^2$$

where

$$\text{D}_\delta(W)=Q\,\text{diag}(\{\max(\sigma_l-\delta,0)\}_{1\le l\le r})V^*.$$

Using Lemma 1, it is easy to obtain the solutions of the minimization subproblems (18)

$$\bar{L}_{i,j}^{k+1,(q)}=D_{\frac{1}{2\mu}}\big(\bar{U}_{i,j}^{(q)}\big),\quad q=1,2,\ldots,p.$$

Then, the $(k+1)$th updating of $\mathcal{L}_{i,j}^{k+1}$ can be obtained via inverse Fourier transform

$$\mathcal{L}_{i,j}^{k+1}=\text{ifft}\big(\bar{\mathcal{L}}_{i,j}^{k+1},[\ ],3\big).\tag{19}$$

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2) With $\mathcal{L}_{i,j}$ fixed, the minimization subproblem for $\mathcal{S}_{i,j}$ can be deduced from (14) as follows:

$$\mathcal{S}_{i,j}^{k+1} = \arg\min_{\mathcal{S}_{i,j}} \lambda\|\mathcal{S}_{i,j}\|_1 + \frac{\mu}{2}\left\|\mathcal{S}_{i,j} - \left(\mathcal{O}_{i,j} - \mathcal{L}_{i,j}^{k+1} + \Lambda_{i,j}^{\mathcal{O}}/\mu\right)\right\|_F^2.$$

The solution of the above optimization subproblem can be directly obtained by the soft threshold

$$\mathcal{S}_{i,j}^{k+1} = \mathrm{Soft}_{\lambda/\mu}\left(\mathcal{O}_{i,j} - \mathcal{L}_{i,j}^{k+1} + \Lambda_{i,j}^{\mathcal{O}}/\mu\right) \tag{20}$$

where

$$\mathrm{Soft}_{\delta}(x) = \begin{cases} x-\delta, & \text{if } x > \delta \\ x+\delta, & \text{if } x < -\delta \\ 0, & \text{otherwise.} \end{cases}$$

2) Global L_{1-2}SSTV Regularized Reconstruction Problem for $(\mathcal{J},\mathcal{X})$: With the other parameters fixed, subproblem (13) for $(\mathcal{J},\mathcal{X})$ can be reformulated as

$$\begin{aligned}&\arg\min_{\mathcal{J},\mathcal{X}} \ell(\mathcal{L}^{k+1},\mathcal{J},\mathcal{X}) \\ &= \sum_{i,j}\left(\langle\Lambda_{i,j}^{\mathcal{L}}, \mathcal{L}_{i,j} - \mathcal{J}_{i,j}\rangle + \frac{\mu}{2}\left\|\mathcal{L}_{i,j} - \mathcal{J}_{i,j}\right\|_F^2\right) \\ &\quad + \langle\Lambda_{\mathcal{X}}, \mathcal{J} - \mathcal{X}\rangle \\ &\quad + \frac{\mu}{2}\|\mathcal{J} - \mathcal{X}\|_F^2 + \tau\|\mathcal{X}\|_{L_{1-2}\mathrm{SSTV}}.\end{aligned} \tag{21}$$

Similarly, we alternately update the two variables $\mathcal{J}$ and $\mathcal{X}$. Then, the optimization problem (21) can be separated into two simpler minimization subproblems.

1) With $\mathcal{X}$ fixed, the minimization subproblem for $\mathcal{J}$ can be deduced from (21) as follows:

$$\arg\min_{\mathcal{J}} \frac{\mu}{2}\|\mathcal{J} - \mathcal{X} + \Lambda_{\mathcal{X}}/\mu\|_2^2 + \sum_{i,j}\left(\frac{\mu}{2}\left\|\mathcal{L}_{i,j} - \mathcal{J}_{i,j} + \Lambda_{i,j}^{\mathcal{L}}/\mu\right\|_F^2\right).$$

This is a convex function, which has the following closed-form solution:

$$\mathcal{J} = \left(\mathcal{X} - \Lambda_{\mathcal{X}}/\mu + \sum_{i,j}\mathcal{P}_{i,j}^T\left(\mathcal{L}_{i,j} + \Lambda_{i,j}^{\mathcal{L}}/\mu\right)\right) \Big/ \left(\mathbf{1} + \sum_{i,j}\mathcal{P}_{i,j}^T\mathcal{P}_{i,j}\right) \tag{22}$$

where the binary operator $\mathcal{P}_{i,j}$ is previously defined, $\mathcal{P}_{i,j}^T$ is its inverse, and $\mathbf{1}$ represents an all-one tensor of size $m \times n \times p$.

2) With $\mathcal{J}$ fixed, the minimization subproblem for $\mathcal{X}$ can be deduced from (21) as follows:

$$\arg\min_{\mathcal{X}} \tau\|\mathcal{X}\|_{L_{1-2}\mathrm{SSTV}} + \frac{\mu}{2}\|\mathcal{X} - (\mathcal{J}^{k+1} + \Lambda_{\mathcal{X}}/\mu)\|_F^2. \tag{23}$$

The DCA method [43] can be used to solve the subproblem (23). Let $\nu = \mu/\tau$; we rewrite the objective function in subproblem (23) as

$$F(\mathcal{X}) := \|\mathcal{X}\|_{\mathrm{SSTV}}^{\mathrm{ani}} - \alpha\|\mathcal{X}\|_{\mathrm{SSTV}}^{\mathrm{iso}} + \frac{\nu}{2}\|\mathcal{X} - (\mathcal{J} + \Lambda_x/\mu)\|_F^2.$$

Then, we decompose $F(\mathcal{X})$ into the difference of two convex components, that is

$$F(\mathcal{X}) = G(\mathcal{X}) - H(\mathcal{X})$$

where

$$\begin{aligned}H(\mathcal{X}) &= \alpha\|\mathcal{X}\|_{\mathrm{SSTV}}^{\mathrm{iso}} + c\|\mathcal{X}\|_F^2 \\ G(\mathcal{X}) &= \|\mathcal{X}\|_{\mathrm{SSTV}}^{\mathrm{ani}} + c\|\mathcal{X}\|_F^2 + \frac{\lambda}{2}\|\mathcal{X} - (\mathcal{J} + \Lambda_{\mathcal{X}}/\mu)\|_F^2\end{aligned}$$

and c is a positive constant to ensure strong convexity of G and H. Starting from an initial point $\mathcal{X}^0$, the DCA iteratively constructs this sequence in the wth iteration

$$\mathcal{X}^{w+1} = \arg\min_{\mathcal{X}} G(\mathcal{X}) - \langle\partial H(\mathcal{X}^w), \mathcal{X}\rangle$$

where $\partial H(\mathcal{X}^w)$ denotes a subgradient of $H(\mathcal{X})$ at $\mathcal{X}^w$.

Let $\tau_x = \tau_y = \tau_z = 1$ in (8). After linearizing the H term, the above iterative sequence is reformulated as

$$\begin{aligned}&\mathcal{X}^{w+1} \\ &= \arg\min_{\mathcal{X}} \|D_x\mathcal{X}\|_1 + \|D_y\mathcal{X}\|_1 + \|D_z\mathcal{X}\|_1 + c\|\mathcal{X}\|_F^2 \\ &\quad - \alpha\langle D\mathcal{X}, h^w\rangle - 2\ c\langle\mathcal{X}, \mathcal{X}^w\rangle + \frac{\nu}{2}\|\mathcal{X} - (\mathcal{J} + \Lambda_{\mathcal{X}}/\mu)\|_F^2\end{aligned} \tag{24}$$

where $D = (D_x, D_y, D_z)$, and $D\mathcal{X} = (D_x\mathcal{X}, D_y\mathcal{X}, D_z\mathcal{X})$

$$h^w = \left(h_x^w, h_y^w, h_z^w\right) = \frac{(D_x\mathcal{X}^w, D_y\mathcal{X}^w, D_z\mathcal{X}^w)}{\sqrt{(D_x\mathcal{X}^w)^2 + (D_y\mathcal{X}^w)^2 + (D_z\mathcal{X}^w)^2}}.$$

The minimization subproblem (24) can be solved by the split Bregman technique [40]. Specifically, we introduce auxiliary variables d_x, d_y, d_z and the Lagrangian multiplier b_x, b_y, b_z. Subproblem (24) can be reformulated as

$$\begin{aligned}\mathcal{X}^{w+1} = \arg\min_{\mathcal{X},d_x,d_y,d_z,b_x,b_y,b_z} &\|d_x\|_1 + \|d_y\|_1 + \|d_z\|_1 \\ &+ c\|\mathcal{X}\|_F^2 - \alpha\left(d_x^T\cdot h_x^w + d_y^T\cdot h_y^w + d_z^T\cdot h_z^w\right) \\ &- 2\ c\langle\mathcal{X}, \mathcal{X}^w\rangle + \frac{\nu}{2}\|\mathcal{X} - (\mathcal{J} + \Lambda_{\mathcal{X}}/\mu)\|_F^2 \\ &+ \frac{\gamma}{2}\|d_x - D_x\mathcal{X} - b_x\|_F^2 + \frac{\gamma}{2}\|d_y - D_y\mathcal{X} - b_y\|_F^2 \\ &+ \frac{\gamma}{2}\|d_z - D_z\mathcal{X} - b_z\|_F^2\end{aligned}$$

where γ is the penalty parameter. It can also be decomposed into three minimization subproblems as follows:

$$\begin{aligned}\mathcal{X}^{w+1} = \arg\min_{\mathcal{X}}\ & c\|\mathcal{X}\|_F^2 - 2\ c\langle\mathcal{X}, \mathcal{X}^w\rangle \\ &+ \frac{\nu}{2}\|\mathcal{X} - (\mathcal{J} + \Lambda_{\mathcal{X}}/\mu)\|_F^2 + \frac{\gamma}{2}\|d_x - D_x\mathcal{X} - b_x\|_F^2 \\ &+ \frac{\gamma}{2}\|d_y - D_y\mathcal{X} - b_y\|_F^2 + \frac{\gamma}{2}\|d_z - D_z\mathcal{X} - b_z\|_F^2\end{aligned} \tag{25}$$

$$\begin{cases} d_x^{w+1} = \arg\min_{d_x} \|d_x\|_1 + \dfrac{\gamma}{2}\|d_x - D_x\mathcal{X}^{w+1} - b_x - \dfrac{\alpha h_x^w}{\gamma}\|_F^2 \\ d_y^{w+1} = \arg\min_{d_y} \|d_y\|_1 + \dfrac{\gamma}{2}\|d_y - D_y\mathcal{X}^{w+1} - b_y - \dfrac{\alpha h_y^w}{\gamma}\|_F^2 \\ d_z^{w+1} = \arg\min_{d_z} \|d_z\|_1 + \dfrac{\gamma}{2}\|d_z - D_z\mathcal{X}^{w+1} - b_z - \dfrac{\alpha h_z^w}{\gamma}\|_F^2 \end{cases} \tag{26}$$

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$$\begin{cases} b_x^{w+1} = b_x + D_x\mathcal{X}^{w+1} - d_x^{w+1} \\ b_y^{w+1} = b_y + D_y\mathcal{X}^{w+1} - d_y^{w+1} \\ b_z^{w+1} = b_z + D_z\mathcal{X}^{w+1} - d_z^{w+1}. \end{cases} \tag{27}$$

The minimizations with respect to $\mathcal{X}$ in (25) and (d_x, d_y, d_z) in (26) have closed-form solutions as follows:

$$\begin{aligned} \mathcal{X}^{w+1} = {} & [(2c+\nu)\mathcal{I} + \gamma\, D^T D]^{-1} \\ & \times \left[2c\mathcal{X}^w + \nu(\mathcal{J} + \Lambda_{\mathcal{X}}/\mu)\right. \\ & \left. + \gamma\, D_x^T(d_x - b_x) + \gamma\, D_y^T(d_y - b_y) + \gamma\, D_z^T(d_z - b_z)\right] \end{aligned} \tag{28}$$

$$\begin{cases} d_x^{w+1} = \text{Soft}\,\dfrac{1}{\gamma}\left(D_x\mathcal{X}^{w+1} + b_x + \dfrac{\alpha h_x^w}{\gamma}\right) \\ d_y^{w+1} = \text{Soft}\,\dfrac{1}{\gamma}\left(D_y\mathcal{X}^{w+1} + b_y + \dfrac{\alpha h_y^w}{\gamma}\right) \\ d_z^{w+1} = \text{Soft}\,\dfrac{1}{\gamma}\left(D_z\mathcal{X}^{w+1} + b_z + \dfrac{\alpha h_z^w}{\gamma}\right). \end{cases} \tag{29}$$

3) Updating the Lagrangian Parameters $\Lambda_{i,j}^{\mathcal{O}}$, $\Lambda_{i,j}^{\mathcal{L}}$ and $\Lambda_{\mathcal{X}}$: Finally, the Lagrangian parameters $\Lambda_{i,j}^{\mathcal{O}}$, $\Lambda_{i,j}^{\mathcal{L}}$ and $\Lambda_{\mathcal{X}}$ can be updated as follows:

$$\begin{cases} \Lambda_{i,j}^{\mathcal{O}} = \Lambda_{i,j}^{\mathcal{O}} + \mu(\mathcal{O}_{i,j} - \mathcal{L}_{i,j} - S_{i,j}) \\ \Lambda_{i,j}^{\mathcal{L}} = \Lambda_{i,j}^{\mathcal{L}} + \mu(\mathcal{L}_{i,j} - \mathcal{J}_{i,j}) \\ \Lambda_{\mathcal{X}} = \Lambda_{\mathcal{X}} + \mu(\mathcal{J} - \mathcal{X}). \end{cases} \tag{30}$$

The main per-iteration cost lies in the update of $\mathcal{L}_{k+1}, \mathcal{X}_{k+1}, \mathcal{S}_{k+1}$ and $\mathcal{J}_{k+1}$. The update of $\mathcal{L}_{k+1}$ requires computing FFT and n_3 SVDs of $n_1 \times n_2$ matrices. The update of $\mathcal{X}_{k+1}$ requires computing DCA for the minimization L_{1-2} norm, which amounts to solving an L_1 minimization problem iteratively as a subproblem. The updates of $\mathcal{S}_{k+1}$ and $\mathcal{J}_{k+1}$ only need to perform basic tensor operations. Thus, the per-iteration complexity is $O(mnp\log(p) + m_{(1)}n_{(1)}^2 p + m_1 n_1 p + mnp + m_1 n_1 n_{(2)} p)$, where $m_{(1)} = \max(m,n)$, $n_{(1)} = \min(m,n)$ and $m_{(2)} = \max(m_1, n_1)$.

Algorithm 1 HSI Restoration via the TLR-L_{1-2}SSTV Model

Input: $m \times n \times p$ observed HSI $\mathcal{O}$, patch size $m_1 \times n_1$, stopping criterion ε, and regularization parameters λ, τ, α.

Output: Denoised image $\mathcal{X}$;

1: Initialize: $\mathcal{L} = \mathcal{X} = \mathcal{S} = \mathcal{J} = 0$, $\Lambda_{i,j}^{\mathcal{O}}, \Lambda_{i,j}^{\mathcal{L}}, \Lambda_{\mathcal{X}} = 0$, $\mu = 10^{-2}$, $\mu_{\max} = 10^6$, $\rho = 1.5$ and $k = 0$;
2: Update all patches $\left(\mathcal{L}_{i,j}, \mathcal{S}_{i,j}\right)$ by (19) and (20) respectively;
3: Update $\mathcal{J}$ by (22);
4: Update $\mathcal{X}$ by the DCA iteration (28), (29) and (27);
5: Update the Lagrangian multipliers by (30);
6: Update the penalty parameter by $\mu := \min(\rho\mu, \mu_{\max})$
7: Check the convergence condition
8: $\max\left\{\left\|\mathcal{O}_{i,j} - \mathcal{L}_{i,j}^{k+1} - \mathcal{S}_{i,j}^{k+1}\right\|_\infty, \left\|\mathcal{J}^{k+1} - \mathcal{X}^{k+1}\right\|_\infty\right\} \le \varepsilon$.

In the TLR-L_{1-2} SSTV restoration algorithm, the inputs include the observed HSI $\mathcal{O} \in \mathbb{R}^{m\times n\times p}$, the stopping criteria ϵ, and the regularized parameters τ, λ, and α. Considering the fact that these parameters have certain proportional relationships, we simply set $\alpha = 0.5$ and tune λ and τ carefully. More details and discussions are presented in Section IV. In addition, for μ, we first initialize it as $\mu = 10^{-2}$ and then update it via $\mu = \min(\rho\mu, \mu_{\max})$, which has been widely used in ALM-based algorithms [44].

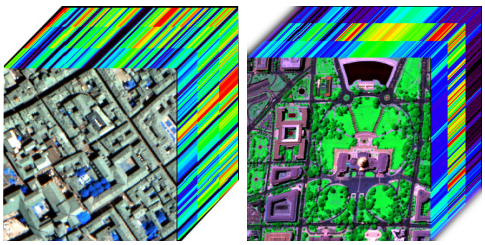

Fig. 3. Data sets used in the simulated experiment. (a) Pavia City Centre data set (R: 20, G: 50, and B: 80). (b) HYDICE Washington DC Mall (R: 50, G: 80, and B: 150).

IV. Experimental Results and Discussion

To verify the effectiveness of our proposed model for HSI restoration, various experiments are performed on a set of challenging simulated and real HSI data. For thorough comparison, nine different state-of-the-art HSI restoration methods are employed as the benchmark in the experiments, L_{1-2}TV [40], LTR-based ITSReg [24], LRTDTV [12], BM3D [9], BM4D [59]; low-rank matrix approximation-based LRMR [14], NAILRMA [15], and LLRSSTV [52]; low-rank matrix factorization-based LRTV [36]. Since the L_{1-2}TV, BM3D, and BM4D methods are only suitable to remove Gaussian noise, we implement them on HSIs that are preprocessed by the RPCA restoration method.

Before performing the restoration model, the gray values in each HSI band are normalized to [0, 1]. After the restoration, each band in the HSIs is converted to the original gray level. The parameter selection in the comparison models is consistent with the description in the original papers. In Section IV-C, the parameters in our proposed TLR-L_{1-2}SSTV model are discussed in detail. To thoroughly evaluate the performance of different restoration methods, a visual comparison and quantitative comparison are adopted to provide quality assessments. Especially for quantitative comparison, five quantitative picture quality indices (PQIs), including the peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and feature similarity (FSIM) [60], erreur relative globale adimensionnelle de synthèse (EGRAS) [61] and spectral angle mapper (SAM) [62] are used to obtain qualitative evaluations of the restoration HSIs. PSNR and SSIM are two conventional PQIs in image processing and computer vision. They evaluate the similarity between the target image and the reference image based on MSE and structural consistency, respectively. FSIM emphasizes the perceived consistency with the reference image. SAM calculates the average angle between spectrum vectors of the denoised HSI and the reference HSI across all spatial positions. The PQI of restored HSIs is calculated band-by-band, and then the PQIs of all bands are averaged to establish the final numerical evaluation criteria: mean PSNR (MPSNR),

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TABLE I

QUANTITATIVE EVALUATION RESULTS OF THE DIFFERENT DENOISING METHODS WITH THE SIMULATED NOISE IN CASES 1–5 ON WASHINGTON DC MALL DATA SET

Noise case	Level	PQIs	ITSReg	BM3D	L_{1-2}TV	LRTV	BM4D	NAI -LRMA	LRMR	LR -TDTV	LLR -SSTV	TLR-L_{1-2}SSTV
case 1	P=0.05 G=0.025	MPSNR	32.730	36.516	32.347	38.645	39.904	31.85	41.327	39.688	40.204	**43.575**
		MSSIM	0.913	0.942	0.857	0.972	0.98	0.879	0.977	0.975	0.982	**0.987**
		MFSIM	0.961	0.961	0.873	0.978	0.986	0.951	0.986	0.986	0.988	**0.993**
		ERGAS	110.753	67.318	107.329	52.332	45.463	117.568	38.366	45.221	43.428	**30.262**
		MSAM	0.079	0.437	0.068	0.027	0.029	0.083	0.026	0.033	0.029	**0.020**
	P=0.1 G=0.05	MPSNR	28.114	32.214	31.390	36.346	35.625	27.024	36.967	36.786	36.771	**39.185**
		MSSIM	0.834	0.872	0.844	0.949	0.952	0.779	0.943	0.520	0.960	**0.965**
		MFSIM	0.934	0.905	0.863	0.96	0.965	0.914	0.969	0.969	0.978	**0.984**
		ERGAS	206.195	106.219	117.764	67.039	71.749	220.895	64.001	62.761	63.77	**48.625**
		MSAM	0.140	0.571	0.076	0.041	0.048	0.150	0.045	0.047	0.044	**0.034**
	P=0.15 G=0.075	MPSNR	24.978	30.142	30.405	34.492	33.053	23.921	33.698	34.759	34.091	**36.36**
		MSSIM	0.758	0.818	0.830	0.921	0.921	0.689	0.896	0.923	0.929	**0.942**
		MFSIM	0.905	0.86	0.853	0.937	0.943	0.877	0.943	0.949	0.962	**0.972**
		ERGAS	312.343	133.195	129.505	83.169	94.83	329.822	93.521	79.843	85.38	**66.222**
		MSAM	0.203	0.647	0.085	0.055	0.064	0.217	0.066	0.061	0.053	**0.049**
	P=0.2 G=0.1	MPSNR	22.642	28.763	29.440	32.902	31.201	21.667	31.251	33.086	31.916	**34.502**
		MSSIM	0.685	0.774	0.814	0.89	0.89	0.612	0.84	0.891	0.888	**0.910**
		MFSIM	0.875	0.823	0.842	0.914	0.92	0.847	0.912	0.927	0.941	**0.958**
		ERGAS	421.760	154.755	142.709	100.57	116.159	439.368	122.694	97.459	107.963	**81.955**
		MSAM	0.263	0.694	0.094	0.066	0.079	0.278	0.086	0.075	0.066	**0.061**
case 2	impulse + Gaussian + deadline	MPSNR	26.355	32.173	29.719	33.931	33.502	25.337	32.458	34.359	33.178	**35.351**
		MSSIM	0.772	0.912	0.823	0.913	**0.94**	0.73	0.869	0.919	0.896	0.937
		MFSIM	0.905	0.936	0.847	0.933	**0.959**	0.893	0.928	0.946	0.946	0.955
		ERGAS	295.026	106.622	139.160	90.447	91.838	309.156	109.163	88.004	94.554	**87.774**
		MSAM	0.193	0.647	0.092	0.081	0.061	0.205	0.083	0.073	0.0100	**0.053**
case 3	impulse + Gaussian + deadline + stripe	MPSNR	26.262	32.105	29.680	33.861	33.418	25.304	32.281	34.308	33.065	**35.227**
		MSSIM	0.767	0.911	0.822	0.912	**0.939**	0.729	0.867	0.918	0.895	0.937
		MFSIM	0.902	0.936	0.846	0.932	**0.958**	0.892	0.927	0.946	0.945	0.954
		ERGAS	296.424	107.282	139.160	91.199	92.62	309.565	110.364	88.688	95.494	**88.508**
		MSAM	0.195	0.648	0.093	0.080	**0.063**	0.205	0.085	0.073	0.102	0.069
case 4	scale coefficient=1	MPSNR	36.126	30.949	32.126	37.128	34.856	40.96	40.507	38.434	40.396	**41.72**
		MSSIM	0.968	0.811	0.853	0.963	0.928	0.984	0.982	0.969	0.983	**0.988**
		MFSIM	0.980	0.851	0.871	0.974	0.947	0.988	0.985	0.982	0.987	**0.991**
		ERGAS	79.236	128.401	111.298	64.961	82.244	41.297	43.205	56.079	42.106	**38.164**
		MSAM	0.049	0.096	0.068	0.040	0.040	0.030	0.034	0.038	0.036	**0.029**
case 5	scale coefficient=2	MPSNR	25.772	30.447	31.327	35.079	33.618	35.688	37.064	36.161	36.879	**37.706**
		MSSIM	0.788	0.8	0.839	0.938	0.916	0.954	0.965	0.949	0.967	**0.971**
		MFSIM	0.911	0.842	0.861	0.957	0.938	0.965	0.974	0.968	0.977	**0.981**
		ERGAS	257.548	136.406	122.673	82.513	94.851	77.33	63.856	72.684	64.564	**61.234**
		MSAM	0.138	0.176	0.076	0.058	0.049	0.054	0.056	0.046	0.066	**0.039**

Note: The bolded value is the optimal value, and the underlined value is the suboptimal value.

mean SSIM (MSSIM), mean FSIM (MFSIM), and mean SAM (MSAM). The higher the values of MPSNR, MFSIM, and MSSIM, and the smaller the values of EGRAS and MSAM; the higher the quality of the image after restoration and the better the restoration method.

A. Simulated HSI Data Experiments

In this section, we select two HSI data sets to apply simulated experiments (see Fig. 3). The first data set is the Pavia City Center [63], which was filmed by the reflection optical system imaging spectrometer (ROSIS-03). The size of the Pavia Center is 1096×1096, with a total of 102 bands. Because some of the bands in the Pavia City Center data set are heavily polluted by noise, they cannot be used as a reference for restoration results. Therefore, this part of the heavily polluted data has been removed. Due to space limitations, we select data with a spatial size of 200×200 and a total of 80 bands for simulated experiments in this part. The second data set is the Washington DC Mall data [64] collected by a HSI acquisition sensor (HYDICE), with panoramic data of 1208×307, covering 191 bands. Due to space limitations, $256 \times 256 \times 191$ subblocks are selected for simulated experiments in this part.

To reduce the computational complexity of the proposed TLR-L_{1-2}SSTV model (10) , the parameters of the local 3-D patches are chosen as follows: the size of 3-D patches from the Pavia Center is set to $m_1 \times n_1 \times p = 60 \times 60 \times 80$, and its step size is set to 20×20; the size of 3-D patches from the Washington DC Mall is set to $m_1 \times n_1 \times p = 80 \times 80 \times 191$, and its step size is set to 40×40.

To simulate noisy HSI data, we add several types of noise to the original HSI data, which can be divided into the following five situations.

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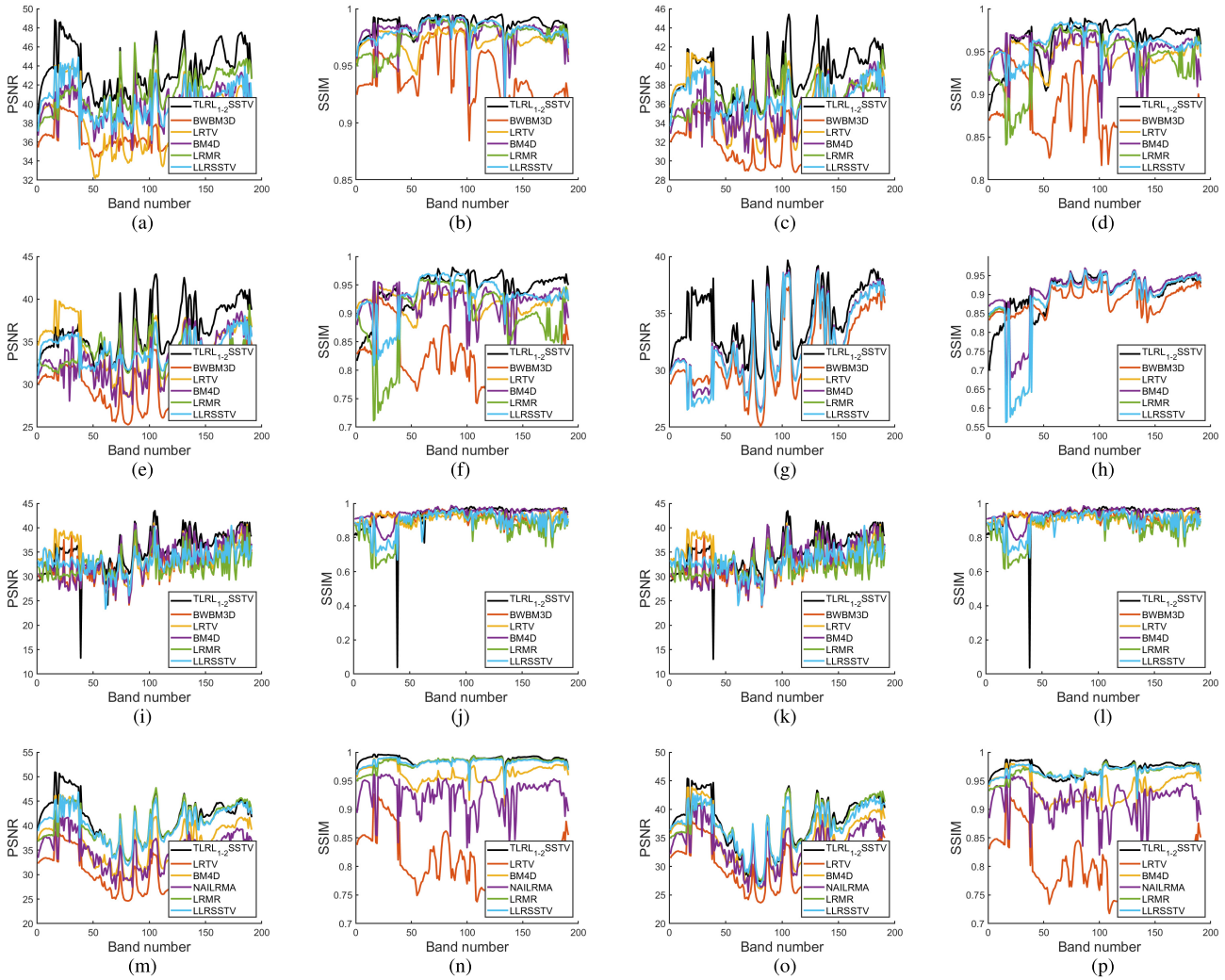

Fig. 4. PSNR and SSIM of all bands for the Washington DC Mall data set. (a) PSNR: P=0.05, G=0.025. (b) SSIM: P=0.05, G=0.025. (c) PSNR: P=0.1, G=0.05. (d) SSIM: P=0.1, G=0.05. (e) PSNR: P=0.15, G=0.075. (f) SSIM: P=0.15, G=0.075. (g) PSNR: P=0.2, G=0.1. (h) SSIM: P=0.2, G=0.1. (i) PSNR: Case 2. (j) SSIM: Case 2. (k) PSNR: Case 3. (l) SSIM: Case 3. (m) PSNR: Case 4. (n) SSIM: Case 4. (o) PSNR: Case 5. (p) SSIM: Case 5.

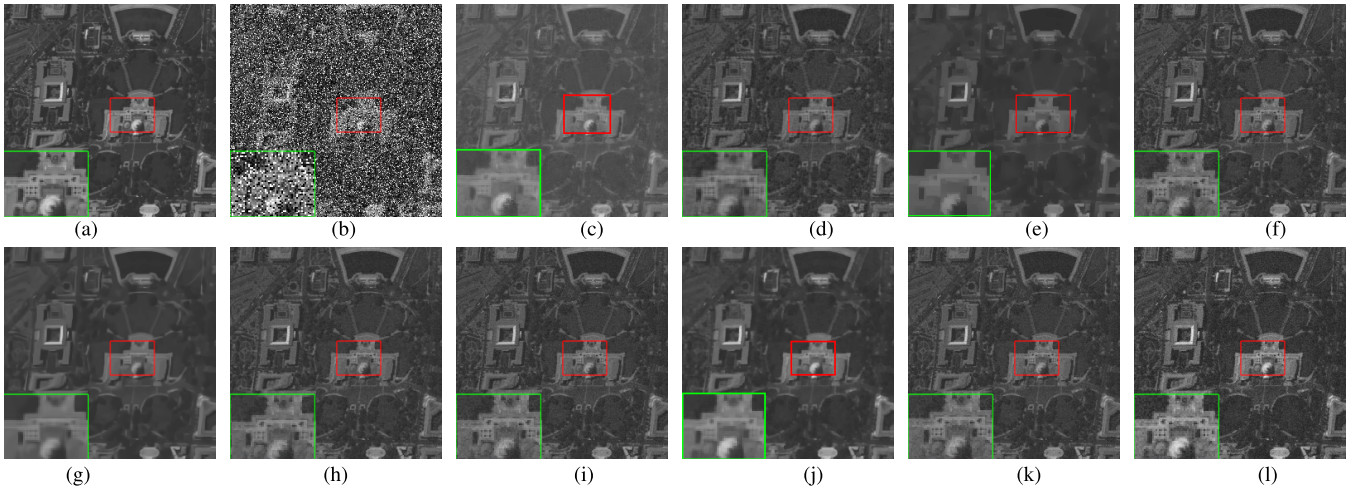

Fig. 5. Denoised results of all methods on band 40 in Washington DC Mall data set, noise case: P=0.2, G=0.1. (a) Original band 40. (b) Noisy image. (c) ITSReg. (d) BM3D. (e) L_{1-2}TV. (f) LRTV. (g) NAILRMA. (h) BM4D. (i) LRMR. (j) LRTDTV. (k) LLRSSTV. (l) TLR-L_{1-2}SSTV.

Case 1: The intensities of noise added in each band are equal. In this case, zero-mean Gaussian noise is added to all bands of the two HSI data. The variances of the Gaussian noise are divided into four levels, denoted as $G = 0.025, 0.050, 0.075, 0.100$. Additionally, pepper-and-salt noise is also added to all bands of the two HSI

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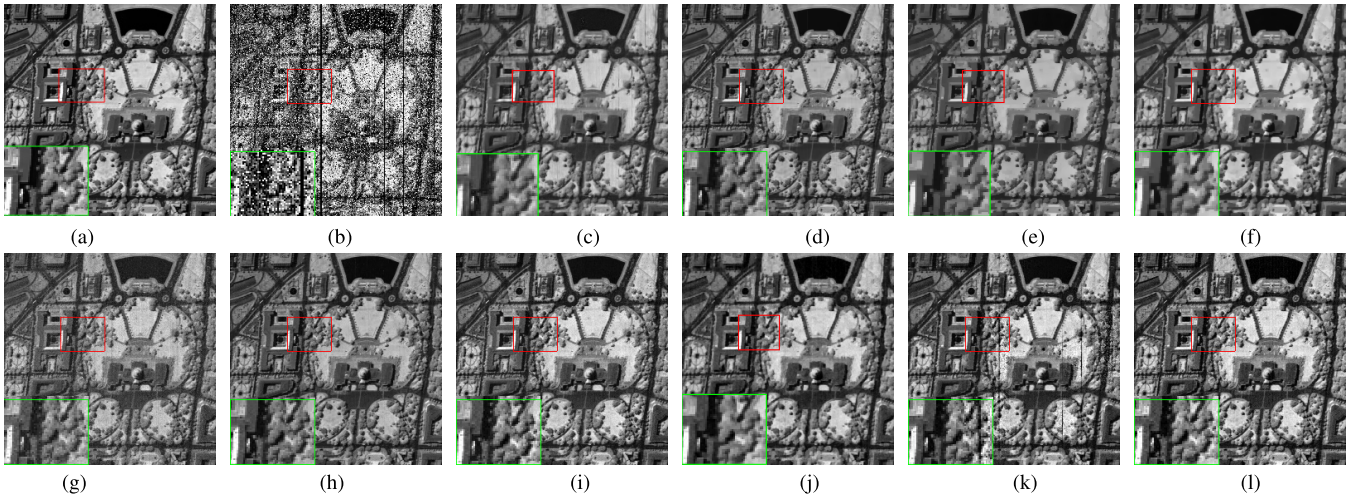

Fig. 6. Denoised results of all methods on band 83 in Washington DC Mall data set in case 3. (a) Original band 83. (b) Noisy image. (c) ITSReg. (d) BM3D. (e) L_{1-2}TV. (f) LRTV. (g) NAILRMA. (h) BM4D. (i) LRMR. (j) LRTDTV. (k) LLRSSTV. (l) TLR-L_{1-2}SSTV.

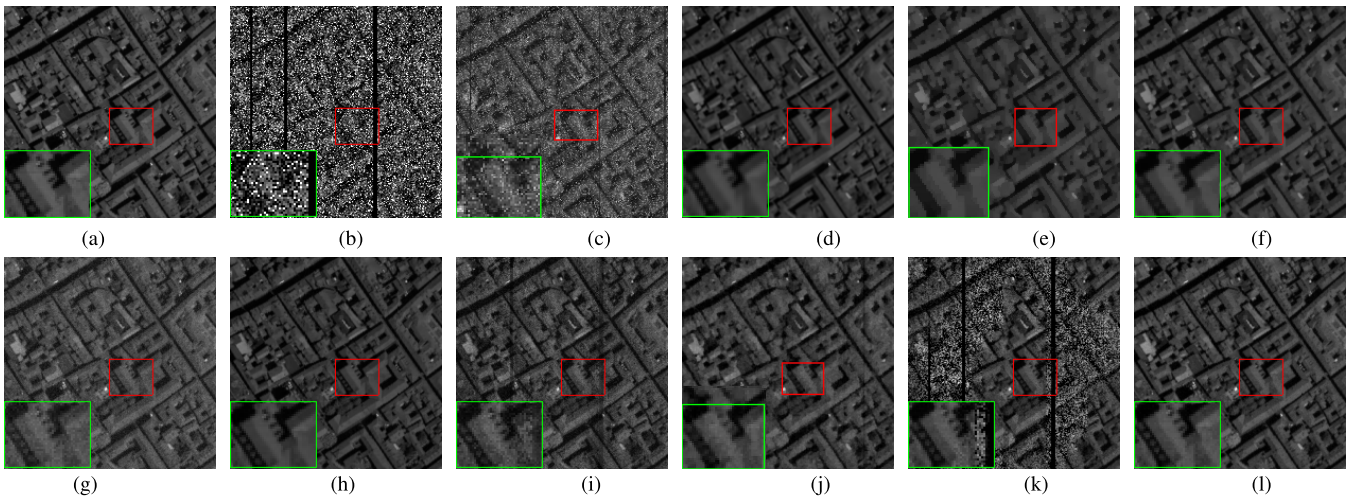

Fig. 7. Denoised results of all methods on band 52 in Pavia City Centre data set in case 3. (a) Original band 52. (b) Noisy image. (c) ITSReg. (d) BM3D. (e) L_{1-2}TV. (f) LRTV. (g) NAILRMA. (h) BM4D. (i) LRMR. (j) LRTDTV. (k) LLRSSTV. (l) TLRL_{1-2}SSTV.

data to simulate sparse noise. The percentages of sparse pepper-and-salt noise are also divided into four levels, denoted as $P = 0.05, 0.10, 0.15, 0.20$.

Case 2: The intensities of noise added in each band are different. In this case, the zero-mean Gaussian noise, pepper-and-salt noise, and dead line noise are added to all bands of the two HSI data. The variance in Gaussian noise and the percentage of pepper-and-salt noise are both randomly selected from 0 to 0.2. In addition, the dead line is added to the 60–90 bands of Washington DC Mall and the 48–58 bands of Pavia City Center. The number of dead lines in the abovementioned bands varies randomly between 3 and 10, and the width of the dead line randomly varies between 1 and 3.

Case 3: On the basis of case 2, stripe noise is added to the 70–100 bands of the Washington DC Mall and the 54–64 bands of the Pavia City Center. The number of stripe noise varies randomly between 20 and 40.

Case 4: In this case, only zero-mean Gaussian noise is added to all bands of the two HSI data. In each band, the variance in Gaussian noise at each pixel is proportional to the gradient magnitude. The scale coefficient at each pixel is set to 1. In other words, the noise variance at each pixel in each band is equal to the gradient magnitude multiplied by 1.

Case 5: On the basis of case 4, the scale coefficient at each pixel is set to 2, that is the noise variance at each pixel in each band is equal to the gradient magnitude multiplied by 2.

1) Washington DC Mall: In this section, for all the noise cases in the Washington DC Mall data set, we list the visual and quantitative quality of all restoration models for comparison.

Table I summarizes the PQIs of restoration results under the condition of the five noise cases in Washington DC Mall. Our proposed method achieves the best PQIs among all the restoration models for comparison in most noise cases. Although the LRMR model also uses patch-based processing for HSI restoration, it lexicographically orders the 3-D patches of HSIs into the 2-D Casorati matrices. Its approach can

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TABLE II

QUANTITATIVE EVALUATION RESULTS OF THE DIFFERENT DENOISING METHODS WITH THE SIMULATED NOISE IN CASES 1–5 ON PAVIA CITY CENTRE DATA SET

Noise case	Level	PQIs	ITSReg	BM3D	L_{1-2}TV	LRTV	BM4D	NAI -LRMA	LRMR	LR TDTV	LLR -SSTV	TLR- L_{1-2}SSTV
case 1	P=0.05 G=0.025	MPSNR	30.911	34.59	30.016	36.368	38.006	31.023	40.743	39.707	40.8	**42.003**
		MSSIM	0.907	0.957	0.868	0.965	0.987	0.914	0.986	0.983	0.987	**0.989**
		MFSIM	0.953	0.975	0.897	0.976	0.991	0.954	0.992	0.991	0.993	**0.994**
		ERGAS	106.495	69.485	115.343	73.061	49.724	104.909	34.372	38.178	34.346	**30.549**
		MSAM	0.114	0.553	0.087	0.087	0.055	0.107	0.059	0.053	0.054	**0.049**
	P=0.1 G=0.05	MPSNR	27.135	30.154	29.097	34.348	34.089	26.82	36.295	36.542	36.579	**37.628**
		MSSIM	0.850	0.885	0.853	0.95	0.967	0.843	0.964	0.965	0.965	**0.969**
		MFSIM	0.927	0.925	0.888	0.965	0.978	0.925	0.98	0.978	0.981	**0.984**
		ERGAS	164.113	114.285	128.404	80.594	75.238	169.632	56.927	54.689	55.354	**52.566**
		MSAM	0.130	0.674	0.097	0.094	0.077	0.130	0.084	0.068	0.075	**0.064**
	P=0.15 G=0.075	MPSNR	24.191	27.958	28.163	32.633	31.513	23.925	33.199	33.208	32.596	**34.323**
		MSSIM	0.784	0.822	0.837	0.929	**0.944**	0.772	0.931	0.932	0.931	0.937
		MFSIM	0.898	0.881	0.877	0.949	0.963	0.898	0.963	0.961	0.964	**0.968**
		ERGAS	230.048	146.654	143.213	92.142	100.021	236.798	80.89	81.440	87.175	**79.495**
		MSAM	0.143	0.723	0.106	0.100	0.093	0.147	0.105	0.082	0.098	**0.080**
	P=0.2 G=0.1	MPSNR	21.903	26.555	27.245	31.077	29.602	21.743	30.914	31.234	30.89	**31.705**
		MSSIM	0.717	0.773	0.818	0.903	**0.919**	0.702	0.892	0.903	0.912	0.892
		MFSIM	0.866	0.846	0.865	0.93	0.948	0.872	0.944	0.941	**0.954**	0.948
		ERGAS	299.268	172.14	159.314	106.523	123.718	304.89	**104.657**	109.582	106.117	106.287
		MSAM	0.157	0.741	0.115	0.107	0.106	0.160	0.121	0.095	0.119	**0.093**
case 2	impulse + Gaussian + deadline	MPSNR	24.781	30.209	27.690	32.474	31.094	25.432	31.261	33.386	32.909	**34.005**
		MSSIM	0.713	0.923	0.828	0.931	**0.952**	0.808	0.897	0.921	0.914	0.946
		MFSIM	0.868	0.954	0.870	0.953	**0.969**	0.909	0.946	0.952	0.953	**0.969**
		ERGAS	255.643	120.747	154.209	95.798	108.187	211.21	107.502	85.807	148.136	**82.098**
		MSAM	0.238	0.634	0.120	0.103	**0.091**	0.164	0.119	0.123	0.167	0.118
case 3	impulse + Gaussian + deadline + stripe	MPSNR	24.789	30.157	27.645	32.4	31.029	25.4	31.142	33.305	32.818	**33.871**
		MSSIM	0.714	0.923	0.827	0.931	**0.952**	0.807	0.896	0.921	0.914	0.945
		MFSIM	0.868	0.953	0.870	0.952	**0.969**	0.908	0.946	0.952	0.953	0.968
		ERGAS	254.995	121.235	154.976	96.281	108.805	211.797	108.906	86.368	151.3	**82.746**
		MSAM	0.237	0.637	0.121	0.103	**0.091**	0.166	0.120	0.125	0.172	0.115
case 4	scale coefficient=1	MPSNR	32.836	28.119	29.343	33.905	33.392	37.383	37.064	36.206	36.989	**37.865**
		MSSIM	0.966	0.808	0.859	0.953	0.955	**0.983**	0.982	0.973	0.981	0.982
		MFSIM	0.981	0.868	0.892	0.969	0.97	0.989	0.989	0.985	0.988	**0.99**
		ERGAS	83.551	143.605	125.169	86.857	81.184	51.128	51.898	56.671	51.991	**48.119**
		MSAM	0.078	0.146	0.090	0.093	0.063	0.067	0.069	0.058	0.056	**0.054**
case 5	scale coefficient=2	MPSNR	22.701	27.425	28.219	31.72	31.12	31.334	32.368	32.201	32.27	**32.818**
		MSSIM	0.765	0.795	0.840	0.929	0.936	0.944	**0.953**	0.947	0.952	0.948
		MFSIM	0.914	0.859	0.881	0.954	0.957	0.966	**0.974**	0.968	0.974	0.969
		ERGAS	268.228	155.933	142.942	102.54	105.565	101.846	88.884	90.014	89.442	**88.026**
		MSAM	0.206	0.289	0.099	0.100	0.082	0.110	0.098	0.074	0.085	**0.069**

Note: The bolded value is the optimal value, and the underlined value is the suboptimal value.

only explore the local low-rank prior in the spectral domain and ignores the spatial prior and the correlations between the small patches. Therefore, the LRMR model obtains the lower PQIs. Based on the LRMA model, the NAILRMA model uses a noise-adjusted iterative strategy and performs well in removing Gaussian noise when the percentage of sparse noise is small. However, when the percentage of sparse noise increases, the efficiency of the NAILRMA model declines rapidly. The LRTV method globally utilizes the low-rank prior in the spectral domain and the sparse prior in the spatial domain simultaneously. The LLRSSTV model locally explores the low-rank prior by the patch-based technique and globally represents the spatial sparsity by the SSTV method. Accordingly, the LRTV model can be viewed as the simplified version of the LLRSSTV model. Hence, they obtain similar PQIs. Compared with the LLRSSTV model, our TLR-L_{1-2}SSTV model improves the SSTV regular term to enhance

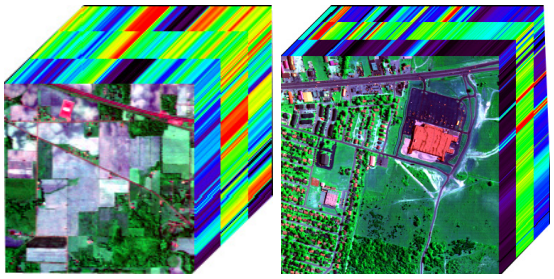

Fig. 8. Data set used in the real data experiment. (Left) AVIRIS Indian Pines data set (R: 1, G: 103, and B: 220). (Right) HYDICE urban data set used in the simulated experiment (R: 20, G: 90, and B: 180).

the ability of sparse representation in the spatial and spectral domains. Additionally, the tensor nuclear norm is used to better approximate the low-rank prior of local 3-D patches

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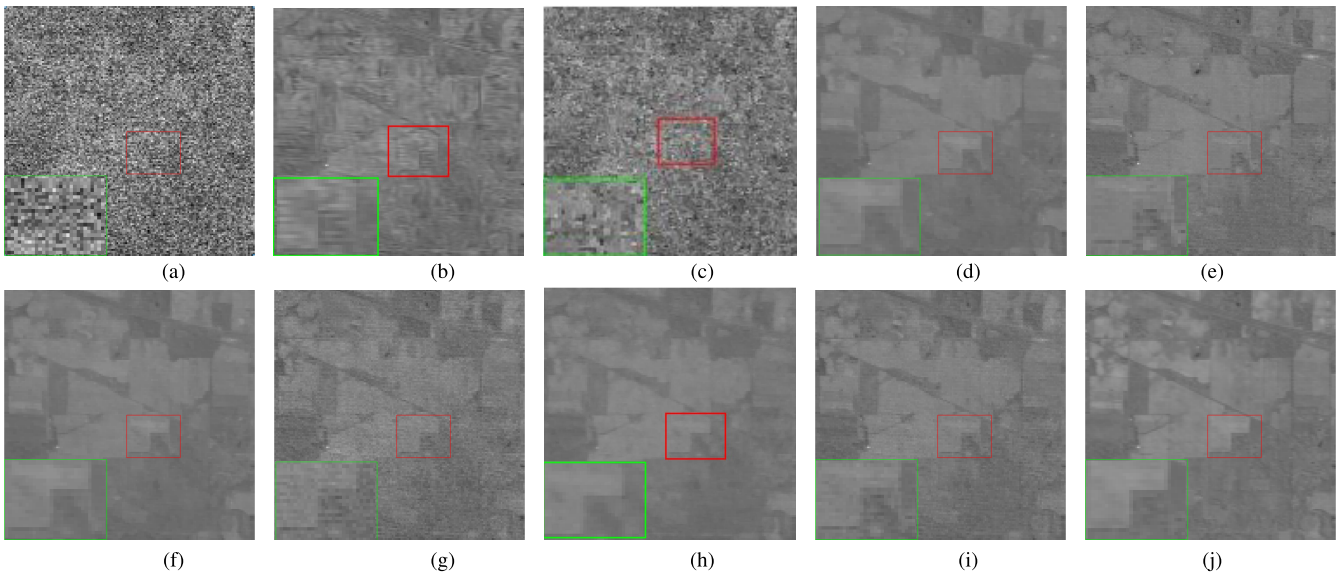

Fig. 9. Denoised results of all methods on band 150 in Indian Pines data set. (a) Original band 150. (b) ITSReg. (c) L_{1-2}TV. (d) LRTV. (e) NAILRMA. (f) BM4D. (g) LRMR. (h) LRTDTV. (i) LLRSSTV. (j) TLRL_{1-2}SSTV.

in HSIs. Therefore, our restoration results obtain better PQIs. Although the 39th band with poor original image quality failed to obtain the ideal PSNR and SSIM, one can find a similar illustration in Fig. 4, which lists the PSNR and SSIM values of restoration results band-by-band in all noise cases.

Under the condition of noise cases 1–4 in the Washington DC Mall data, Fig. 5 shows the grayscale images of the 40th band in different restoration results. Under the condition of the more complex noise case, for example, noise case 3, which includes Gaussian noise, sparse noise, dead line noise, and stripe noise with different noise intensities in each band, Fig. 6 lists the grayscale images of the 83rd band in different restoration results. Figs. 5(a) and 6(a) are original images in the 40th and 83rd bands, and Figs. 5(b) and 6(b) are their noise versions where the detailed information can hardly be distinguished due to the high noise intensity. According to the partial enlargements in Figs. 5 and 6, one can obviously find the following facts: the restoration results of both the NAILRMA and LRMR models have Gaussian noise residues, and the results of the LRTV model lose the image details due to the global low-rank constraint. Because of the local low-rank prior and global L_{1-2}SSTV regularization in both the spatial and spectral domains, our proposed model obtains a better visual effect and restores more details and texture information in visual comparison with other models.

2) Pavia City Center: To further verify the versatility of our model for different data sets, in this part, we perform the same experiments on the Pavia City Center data set as the Washington DC Mall data set under the condition of the five noise cases. Table II summarizes the PQIs of restoration results under the condition of the five noise cases in the Pavia City Center data set. One can clearly find that our proposed model achieves significantly higher MPSNR values than those of the comparison models. For visual comparison, in noise case 3, Fig. 7 lists the grayscale images of the 52nd band in the original HSI data, the noise data, and the different restoration results. It can also be seen that our model better preserves the local details and texture information while effectively removing the complex noises.

B. Real HSI Data Experiments

In this section, two HSI data sets with real noise are adopted in our experiments. They are the Indian Pines data set and the HYDICE Urban data set in Fig. 8.

1) Indian Pines: The Indian Pines data set [64] is collected by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). The size of the space is 145×145, with a total of 220 bands. Some bands of the data set are seriously polluted by impulse noise and Gaussian white noise, while some bands preserve high image quality. Here, we set the size of patch $m_1 \times n_1 \times p = 40 \times 40 \times 220$, and the step size of the patch is set to 20×20. For the Indian Pines data set, the restoration is mainly based on the removal of noise in noise bands and the retention of image details in high-quality bands. Fig. 9 lists the grayscale images of the 150th band in the noise HSI and the different restoration results. Fig. 10 shows the vertical mean profiles of the 150th band. In Fig. 10(a), the vertical mean profile of Fig. 9(a) fluctuates sharply, indicating that there is severe noise in the image. The LRMR method removes noise to some extent, but as shown in Figs. 9(f) and 10(f), the effect is not outstanding. The LRTV method can achieve smooth results in Figs. 9(c) and 10(c), but some local fluctuations in the profile are still dramatic. Similar results can also be seen in Figs. 9(g) and 10(g). In contrast, the TLR-L_{1-2}SSTV method can obtain the best results of visual effects in Fig. 9(h) and the smoothest profile in Fig. 10(h).

2) HYDICE Urban: The original size of the HYDICE Urban HSI data set [65] is $207 \times 207 \times 210$. There are 189 bands left after the water absorption band is excluded. The right image in Fig. 8 is a 3-D representation of hyperspectral imagery with false colors synthesized by three bands 20, 90,

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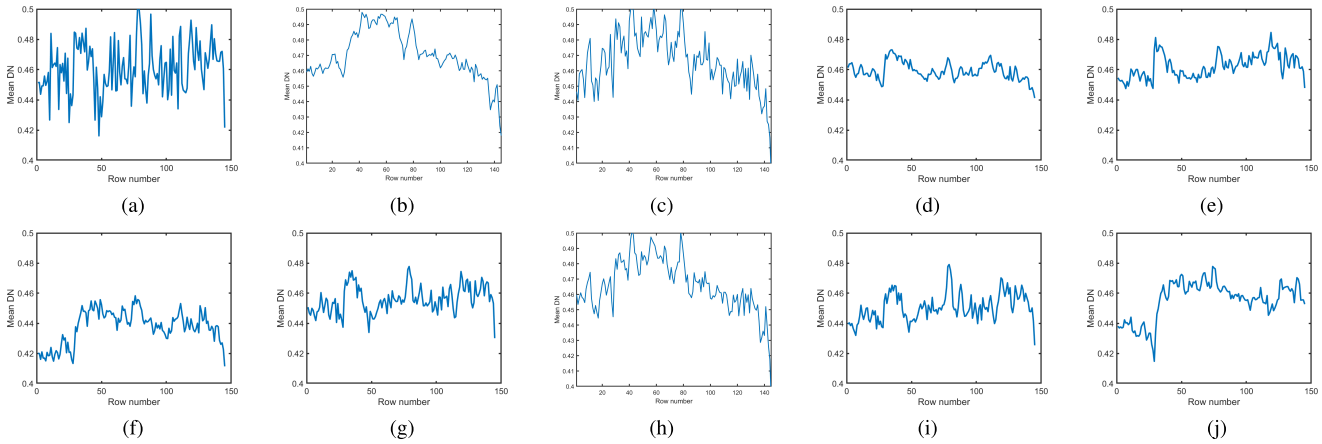

Fig. 10. Vertical mean profiles of band 150 by all the compared methods for the Indian pines. (a) Original band 150. (b) ITSReg. (c) L_{1-2}TV. (d) LRTV. (e) NAILRMA. (f) BM4D. (g) LRMR. (h) LRTDTV. (i) LLRSSTV. (j) TLRL_{1-2}SSTV.

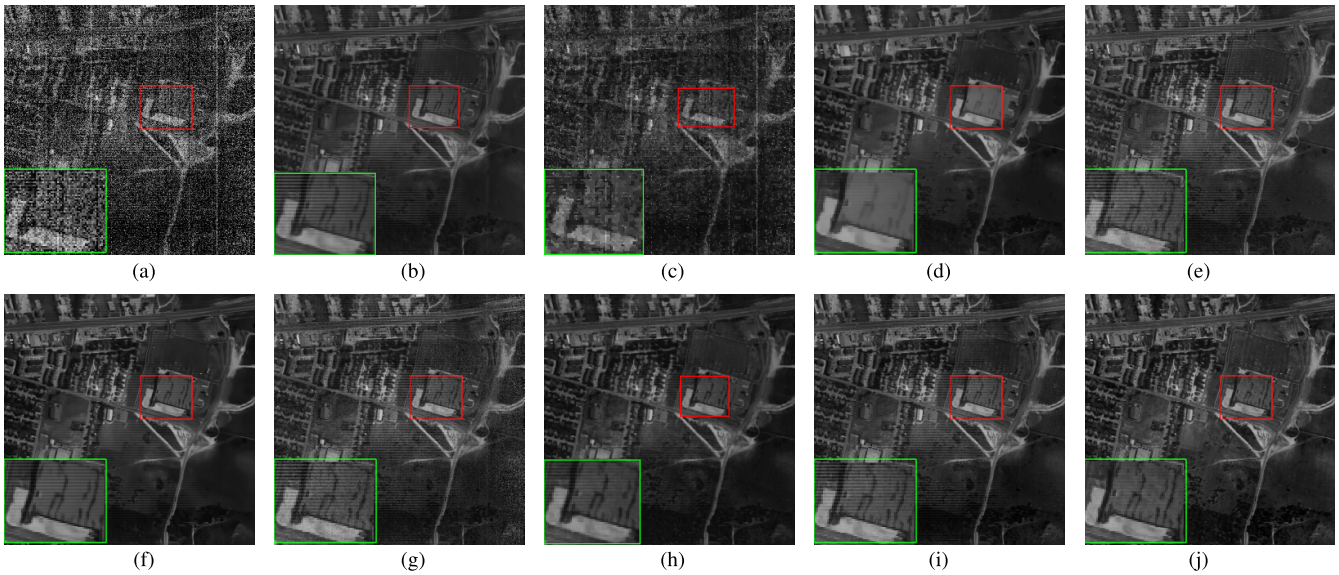

Fig. 11. Denoised results of all methods on band 207 in HYDICE Urban data set. (a) Original band 207. (b) ITSReg. (c) L_{1-2}TV. (d) LRTV. (e) NAILRMA. (f) BM4D. (g) LRMR. (h) LRTDTV. (i) LLRSSTV. (j) TLRL_{1-2}SSTV.

and 180. Here, we set the size of patch $m_1 \times n_1 \times p = 80 \times 80 \times 210$, and the step size of the patch is set to 40×40. Fig. 11 lists the grayscale images of the 207th band in the noise HSI and the different restoration results. One can obviously see that the TLR-L_{1-2}SSTV method restores more spatial details than other methods and removes more sparse noise, especially strip noise. Hence, our method obtains the restoration image with the best visual effect. To further demonstrate the good effect of our method on strip noise removal, in Fig. 12, we present the mean profile curves of the 207th band in the noise HSI and the different restoration results. When the mean profile curve fluctuates significantly, it is suggested that there is banding noise. After restoration by various methods, the fluctuation of the mean value profile of the denoising image is alleviated. As seen from Fig. 12, after denoising, the stripe noise in the results of the LRMR, LLRSSTV, and BM4D methods still exists, and the fluctuation in the result of the TLR-L_{1-2}SSTV method is very small. This phenomenon further reveals that the TLR-L_{1-2}SSTV method has an irreplaceable role in removing hybrid noise.

C. Discussion

1) Sensitivity Analysis of Parameter λ, τ: It is easy to see that λ is the parameter used to restrict the sparsity of the sparse noise, while τ is used to adjust the effect of the L_{1-2}SSTV term. As the analysis stated in the TRPCA model [27], $\lambda = 1/(\max(m, n)p)^{1/2}$ is good enough to guarantee the exact recovery. Owing to the new TV penalty, our model is different from TRPCA. We thus set $\lambda = 1/(\max(m, n)p)^{1/2}$ as the initial value, and adjust the parameter by the specific experimental data set based on this initial value. Under the conditions of the Washington DC and Pavia HSI data sets, Figs. 13 and 14 report the MPSNR and MSSIM values of our TLR-L_{1-2}SSTV model according to the different parameters λ and τ. Therefore, in the simulated and real data experiments, we suggest the use of $\lambda = 0.25$, $\tau = 0.01$.

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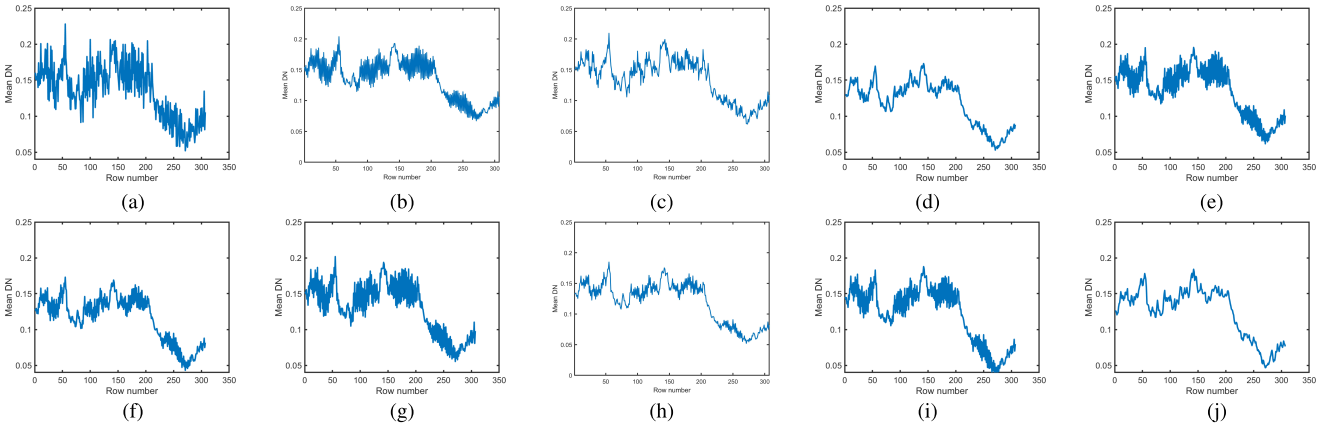

Fig. 12. Vertical mean profiles of band 207 by all the compared methods for the HYDICE Urban. (a) Original band 207. (b) ITSReg. (c) L_{1-2}TV. (d) LRTV. (e) NAILRMA. (f) BM4D. (g) LRMR. (h) LRTDTV. (i) LLRSSTV. (j) TLRL_{1-2}SSTV.

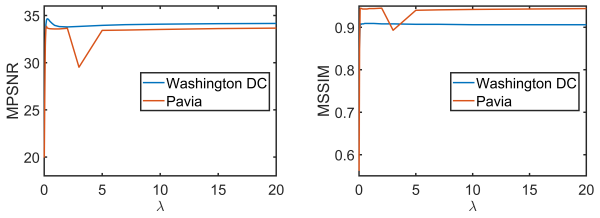

Fig. 13. MPSNR and MSSIM value versus the λ value of TLRL_{1-2}SSTV. (a) Change in the MPSNR value. (b) Change in the MSSIM value.

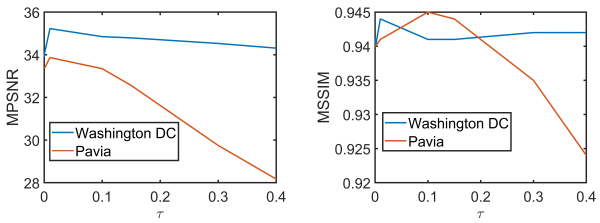

Fig. 14. MPSNR and MSSIM value versus the τ value of TLRL_{1-2}SSTV. (a) Change in the MPSNR value. (b) Change in the MSSIM value.

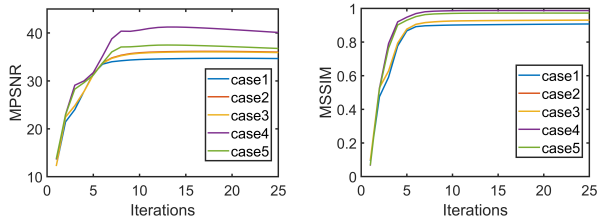

Fig. 15. MPSNR and MSSIM value versus the iteration number of TLRL_{1-2}SSTV. (a) Change in the MPSNR value. (b) Change in the MSSIM value.

2) Convergence Analysis of the Model: Fig. 15 shows the MPSNR value and the MSSIM value of our TLR-L_{1-2}SSTV method according to the iterations. After a few iterations, the MPSNR value and the MSSIM value tend to be stable, which reflects the convergence of our algorithm.

V. Conclusion

Different types of noise have different statistical characteristics, and the existence of mixed noise also greatly limits the subsequent processing and application accuracy of HSI. Aiming at the HSI restoration problem, based on the low-rank and sparse prior of HSI, this article establishes the L_{1-2} SSTV regularization with a stronger sparse representation ability, and combines it with the local low-rank prior of tensors, to propose an HSI restoration model based on local tensor nuclear norm minimization and global L_{1-2}SSTV regularization. Our model not only effectively removes Gaussian noise, impulse noise, deadlines, strips, and their mixed noise, but also reduces the dependence on noise independent and identical distribution assumptions and can partially suppress structure-related noise. The results of a large number of experimental simulations on simulated and real HSI data show that both the quality evaluation index and visual effect of our model restoration results are excellent.

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